

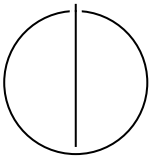
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
FACULTY OF INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — FACULTY OF INFORMATICS

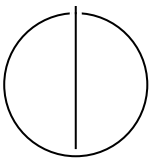
TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

Author:	Lukas Alex Jochim
Examiner:	Prof. Dr. Björn Schuller
Supervisor:	MSc. Simon Pistrosch
Submission Date:	01.09.2025



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

First of all, I want to thank my supervisor, Simon, for guiding me through this thesis by not only giving me valuable feedback but also by providing helpful insights to bring the best out of me and this study. I am also very grateful for my friends who have motivated me throughout the process of this thesis and have always been by my side. Last but not least, I want to express my deepest gratitude and appreciation to my parents and family, who have always supported and believed in me. Without all of you collectively, this thesis would not have been possible.

Transparency in the Use of AI Tools

While writing this thesis, Grammarly was utilised to ensure high-quality grammar, style, clarity, and coherence in the content of each chapter. Furthermore, DeepL was used for translation purposes.

During the development phase, ChatGPT and Claude Sonnet were used to generate simple code snippets and to gather first ideas for implementing certain functionalities. However, everything generated by one of those tools has been carefully evaluated by me to ensure correctness.

Abstract

Music plays a key role in everyday life, being entertaining and engaging complex cognitive and emotional processes in the human brain. Nonetheless, the neural mechanisms behind music perception remain unknown. Therefore, this thesis explores whether Electroencephalography (EEG) signals can be used to reliably decode music perception, aiming to bridge the gap between Music Information Retrieval (MIR) and neurosciences. A dedicated dataset for EEG based Music Genre classification is created to contribute to this area, named the Multimodal Signals & EEG for Classification of Genres (MuSiCGen) dataset. By utilising the GTZAN music dataset, EEG signals have been recorded by a multimodal biosignal headset while listening to music. The recorded signal then underwent a custom denoising process to eliminate most artifacts, followed by an exhaustive feature extraction combined with specialised feature selection methods. Based on these parameters, machine learning classifiers are trained to predict the corresponding genres, which are then analysed regarding accuracy, F1-score, and their confusion matrices. To compare with a state-of-the-art EEG dataset, the same approach is followed with the Database for Emotion Analysis using Physiological Signals (DEAP), which is evaluated in the same way as the newly introduced dataset. Results demonstrate that genre-specific neural patterns can be detected above chance level. Concreteley, mean accuracies of 68.3% and 20.2% were measured for binary classification on the DEAP dataset and ten-class classification on the MuSiCGen dataset respectively. Although signal quality and device limitations posed a challenge, the key findings show promising evidence that EEG signals possess meaningful information about musical perception. Overall, this thesis lays the groundwork for further research in the area of EEG-based MIR by providing a novel database alongside detailed methodology for Music Genre Recognition (MGR) using EEG signals. Future studies can build upon this thesis by employing larger subject pools, advanced deep learning architectures, or multimodal data fusion.

Contents

Acknowledgments **iv**

Transparency in the Use of AI Tools **v**

Abstract **vi**

1 Introduction **1**
1.1 Motivation . . . 1
1.2 Research Goals . . . 2
1.3 Outline . . . 2

2 Theoretical Background **4**
2.1 Medical Foundations of EEG Signals . . . 4
2.2 Brain Rhythms . . . 5
2.3 Recording and Measurement of EEG Signals . . . 6
2.4 Denoising and Preprocessing of EEG signals . . . 8
2.4.1 Types of Artifacts in EEG signals . . . 8
2.4.2 Artifact Removal Techniques . . . 9
2.5 Feature Extraction of EEG Signals . . . 11
2.5.1 Time Domain Feature Extraction . . . 12
2.5.2 Frequency Domain Feature Extraction . . . 13
2.5.3 Time-Frequency Domain Feature Extraction . . . 14
2.5.4 Feature Selection . . . 16

3 Related Work **17**
3.1 Music Genre Classification with Audio Features . . . 17
3.2 EEG Datasets and Studies with Music Stimulus . . . 18
3.2.1 DEAP . . . 19
3.2.2 Music BCI . . . 20
3.2.3 MUSIN-G . . . 20
3.3 Music Genre Recognition with EEG Signals . . . 21
3.4 Emotion Recognition with EEG signals . . . 22

4 Music Datasets **25**
4.1 Audio Set . . . 25
4.2 Million Song Dataset . . . 26
4.3 MagnaTagATune . . . 27
4.4 FMA Dataset . . . 27
4.5 GTZAN Genre classification . . . 28
4.6 Final Decision . . . 29

5 Methodology **31**
5.1 Acquisition . . . 31
5.1.1 General setup, environment and stimuli . . . 31
5.1.2 Acquisition device . . . 32
5.1.3 Technical setup . . . 32
5.2 Denoising and Preprocessing . . . 33
5.3 Feature extraction . . . 36
5.4 Classification . . . 39
5.4.1 DEAP . . . 40
5.4.2 MuSiCGen . . . 42

6 Results **43**
6.1 DEAP . . . 43
6.2 MuSiCGen . . . 45

7 Conclusion **53**
7.1 Key Findings and Results . . . 53
7.2 Limitations and Future Work . . . 53

Abbreviations **55**

List of Figures **58**

List of Tables **59**

Bibliography **60**

1 Introduction

1.1 Motivation

The human brain is one of the most complex and important systems within the human body. It manages various functionalities like emotion, memory, attention and sensory processing. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music serves as a universal tool to influence one's emotions, concentration and cognition. Moreover, the brain gives the unique ability to retrieve many different features out of music, like melodies, the identification of an artist or a specific genre, reacting differently depending on the style of music. However, although the brain is central to cognitive processes such as listening to music, the core mechanisms of music perception remain unknown and are currently black boxes.

To enlighten these black boxes, medical advancements like functional Magnetic resonsance imaging (fMRI) or Electroencephalography (EEG) arrived, which can measure the brain activity. Especially EEG has aroused high interest since it directly records the neural activity through electrical signals and can be done non-invasively. This is why combining EEG and music has the ability to help better understand how the brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and computer science.

There is a growing research interest in the affective influence of music on the brain. However, classifying different music genres based on EEG signals remains underrepresented in current studies. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science. Especially with the growing investigations on Brain Machine Interfaces (BMIs), adaptive music therapy or neurofeedback systems may become possible. Additionally, entertainment technologies can become more user-centred, facilitating unique experiences that impact many sectors like education and mental health.

1.2 Research Goals

The main goal of this thesis is to close the existing research gap between Music Genre Recognition (MGR) and neurosciences to provide a basis for further research in this area. The first goal revolves around providing a dedicated dataset combining both MGR and EEG. As no such dataset currently exists, it is important to develop a sophisticated experiment setup to create a high-quality source of data for MGR tasks. This is why this thesis introduces a novel EEG dataset called the *Multimodal Signals & EEG for Classification of Genres (MuSiCGen) dataset*.

Hand in hand with creating this specialised dataset, this study also explores using unconventional methods to record the EEG signals. In searching for EEG devices suitable for everyday life, it is essential to investigate how well such systems work in practice. For this purpose, a multimodal headset measuring multiple biosignals, including EEG while playing music, is used as an acquisition device to find out how suitable similar headsets are for further research.

Since such multimodal devices have not seen any applications in research yet, another goal is to find a general denoising pipeline to identify recorded signal artifacts. As most publicly available tools for noise reduction in EEG rely on a standardised placement of measurement units, custom methods must be found for further analysis.

Moreover, the most important goal is to find out whether it is possible to infer the musical genre listened to based on neural activity (namely EEG). Since it is unclear how well the multimodal headset can capture EEG signals, a second EEG dataset, called the Database for Emotion Analysis using Physiological Signals (DEAP), is taken into account, on which MGR is performed as well. This way, whether musical genre-specific patterns can be derived from EEG signals becomes independent from the acquisition device, allowing for a more generalised answer to this question.

1.3 Outline

To answer the research questions mentioned above, it is vital to have a straightforward research approach, depicted by the structure of this thesis. A short thesis outline is given in the following, focusing on structure and content in the succeeding chapters.

At first, a theoretical background on EEG signals is given in Chapter 2, building the basis of understanding for all further chapters. After covering the medical basics of EEG signals, the focus moves towards the three key steps when planning an experiment involving EEG with classification as a background. This includes the recording and measurement, denoising techniques, and feature extraction/selection methods for EEG signals.

Chapter 3 then examines the current state of the art in research regarding MGR both on audio and EEG signals, highlighting features and classification methods used in those. With the same focus, emotion recognition using EEG is highlighted as a second classification use case apart from MGR. Besides classication tasks, popular EEG datasets using music as a stimulus are presented as a reference to the newly introduced MuSiCGen dataset.

Following Chapter 3, Chapter 4 discusses different music datasets that can be used for the purposes of this study. At first, the main requirements for a music dataset suitable for this thesis are explained and clarified. Based on these requirements, five different song databases are presented and evaluated against the mentioned requirements, followed by a final decision on which dataset is employed for this thesis.

After that, Chapter 5 describes the methodology behind the study. This includes the recording of the MuSiCGen dataset, followed by denoising and feature extraction steps on the acquired data. Lastly, the approach for MGR on the DEAP dataset, as well as the MuSiCGen is explained.

Chapter 6 then discusses the results, also regarding the goals of this study. This chapter evaluates the classification performance on both datasets, including various metrics that assess how well MGR works on EEG data.

Last but not least, Chapter 7 concludes the thesis as a whole. It not only sums up the thesis's key findings but also mentions problems and limitations, proposing future research based on this study and indicated challenges.

2 Theoretical Background

2.1 Medical Foundations of EEG Signals

Before talking about how Electroencephalography (EEG) signals can be used for clasification or fourther analysis, it is vital to understand how it is created and in what way it can depict the nerual activity.

The Central Nervous system (CNS) of nerve and glia cells can be found between neurons. Nerve cells can recognize and transmit stimuli through other nerve cells to further locations. To understand how this is possible, the structure and functionality of a single nerve cell have to be understood. This kind of cell consists of three main components. Firstly, there is the cell body with a single nucleus. It is responsible for the cell's metabolism and protein synthesis. Secondly, the Axon, a long cylinder, transmits electrical impulses up to a metre through the body. The third component, dendrites, further supports the transportation of electricity. They are either connected to other dendrites or axons of neighboring cells from which they can receive impulses. Vice versa, it is also possible to transmit the impulse to another cell [28].

Related to the transmission of electrical impulses, the so-called Action potential (AP) is very important regarding EEG, since the AP represents information sent by a nerve. In simple words, AP is the temporary change in the membrane potential that is transmitted along the axon caused by the exchange of ions. It lasts between 5 and 10 ms, resulting in a peak during the positive polarization of the membrane. After repolarizing to a negative potential, the nerve returns to resting potential so that it can be excited again. It is important to remember that an AP is only generated if there is a strong enough stimulus. If the stimulus is not powerful enough, only a small local electrical impulse is created, which is not sufficient to be transmitted to another nerve, thus not resulting in an AP [28].

As already mentioned, the AP is a key player when it comes to EEG because an EEG signal indirectly records current flows during the excitation of the synapses. During the transmission of an electric impulse because of the activation of neurons, an electrical field is created next to a magnetic field, which can be measured by specialized EEG systems over the scalp [1]. Since the brain and the scalp consist of multiple layers, only a

[1] cf. Section 2.3

large population of active neurons can be recognized by systems measuring from the scalp hence they have to be amplified for visual inspection [28].

The measured signal can differ based on the patient's mental or emotional state. Therefore, the signal has to be distinguished into certain components to make analysis easier.

2.2 Brain Rhythms

Traditionally, visual inspection has been the standard method used to analyze EEG signals for signs of brain disorders or other abnormalities. Specific amplitudes and frequencies of these signals change in a healthy human depending on their state. For instance, the EEG signal differs between wakefulness and sleep. To distinguish the different appearances of the signal, the frequency range is taken into account to form five brain rhythms, which are shown in Figure 2.1, each of which defining a particular state of the brain [28].

Delta waves occupy the range of 0.5-4 Hz and represent the brain's grey matter [20]. They mainly occur in deep sleep and may be present in the waking state. These signals are very easy to confuse with large muscle artifacts from the jaw or neck, since they are close to the skin's surface. Additionally, the Delta signals coming from the brain are highly attenuated in passing through the skull [28].

Theta waves lie between 4-8 Hz and are primarily present in the thalamic region. They appear when a human's mental state transitions from consciousness towards drowsiness, which is why they dominate during meditation therapy [28, 20]. Combined with that, it also stands for human growth as they are linked to serotonin hormones [20].

Within the range of 8-13 Hz, Alpha signals are present and are shown as a round-shaped signal. Being the most dominant rhythm in the whole brain and standing for the brain's white matter, it indicates relaxed awareness without any attention and concentration [28, 20]. Studies also show that Alpha waves are a waiting pattern by the visual region of the brain since they are reduced or even eliminated by opening the eyes [28].

Signals with frequencies between 14-26 Hz [28] or 13-30 Hz [20] are classified as Beta waves, which stand for the usual waking state of the brain. Therefore, it is associated with active thinking or problem solving [28]. By inducing the production of Cortisol, it is also responsible for learning and memory [20].

Everything above 30 Hz (up to 45 Hz) is considered as Gamma waves. They are very rare in occurrence and are primarily used for detecting and confirming brain diseases [28]. However, they also appear during hyper alertness and the integration of sensory

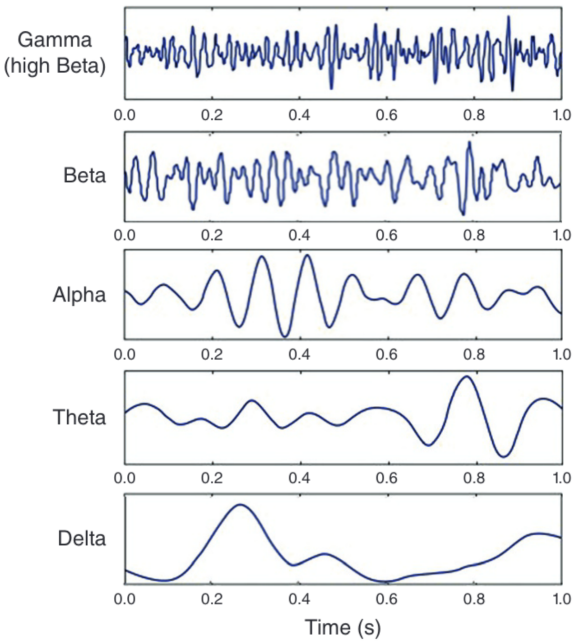


Figure 2.1: The five brain rhythms in EEG [28]

inputs, which is why they combine senses and memory [20].

2.3 Recording and Measurement of EEG Signals

In the previous two sections, it became clear that the electrical activity coming from APs can be used to depict the neural state of the brain. Therefore, covering how these impulses can be measured is also important.

An EEG system mainly consists of multiple electrodes (called channels), differential amplifiers for each channel, and filters. Analogue to Digital converters (ADCs) are used as well to digitise the measured signals, enabling their processing by computerized systems. [28].

The most crucial components for data quality are the electrodes. There are different types of electrodes depending on the study's use case (disposable, reusable disc electrodes, headbands, etc.). However, headbands with Ag-AgCl discs are the most popular choice for multichannel measurements. As already mentioned in Section 2.1, because of the layered anatomy of the scalp, there is high impedance between the cortex and the electrodes, which can lead to deformations in the signal. Therefore, a rule of

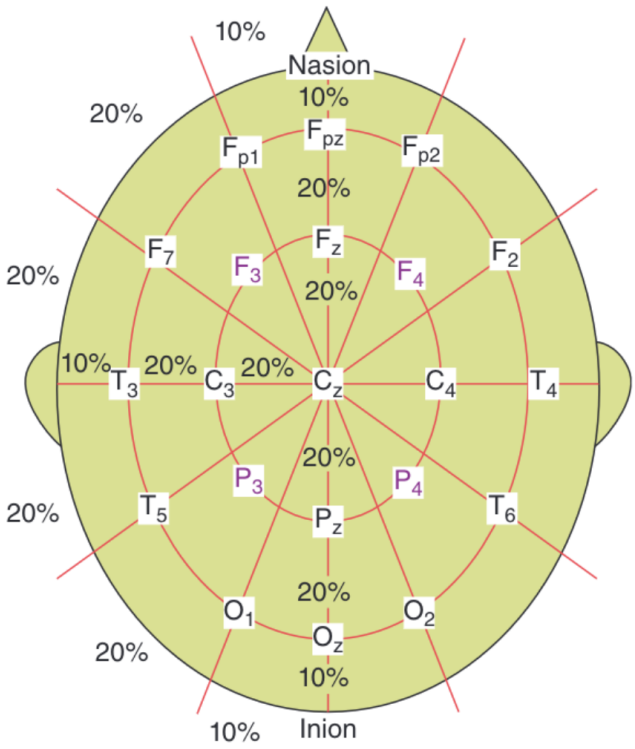


Figure 2.2: Conventional 10-20 EEG electrode positions for 21 electrodes [28]

thumb is that the channel's impedances should be less than $5\,k\Omega$ and balanced to $1\,k\Omega$ within each other [28].

For unification in measurement and placement in multichannel EEG recordings, the 10-20 system is recommended as a conventional electrode setting. A setup for 21 electrodes is illustrated in Figure 2.2. This system avoids eyeball placement and employs constant distances between all electrodes. It is called the 10-20 system because the electrodes are placed at intervals of 10 or 20% of the total front-back/left-right distance of the skull. In the case where more than 21 electrodes are employed, the additional channels would be placed between the existing electrodes with equidistance between each other [28].

As the adequate bandwidth of EEG signals lies around 100 Hz, the system's minimum sample rate should be 200 Hz to meet the Nyquist criterion[2] Depending on the required resolution, sampling frequencies up to 2000 Hz are possible. In quantification of

[2]The Nyquist criterion states that to accurately reconstruct a signal without aliasing, it must be sampled at a rate at least twice its highest frequency component.

the signal, 16 bits for each sample is sufficient and popular for many EEG recording systems [28].

2.4 Denoising and Preprocessing of EEG signals

After acquiring the EEG signals, further steps have to be followed before performing a classification task. Since EEG is a non-invasive technique, other signals, so-called artifacts, are also measured. Different artifacts are presented before discussing possible methods to remove the undesired data.

2.4.1 Types of Artifacts in EEG signals

Physiological Artifacts	**Non-Physiological Artifacts**
Ocular-related artifacts (eye movements and blinking)	Power line interference
Electromyography (EMG) artifacts from head movements	Poor electrode placement on scalp
Electrocardiography (ECG) artifacts from heart beat	Malfunction of EEG recording system components
Scalp perspiration and movements	Digital artifacts from loose wiring or circuit board connections

Table 2.1: Types of artifacts in EEG signals [17]

Before illustrating what types of artifacts exist, it is essential to mention that artifacts themselves do not destroy EEG signals but add to the neural activity. Therefore, the types and characteristics must be explained thoroughly before removing any of this data. In general, artifacts can be put into two groups: the *Non-Physiological artifacts* and the *Physiological artifacts*, further emphasised by Table 2.1. As the names already suggest, while non-physiological artifacts are distortions from the outside world, physiological artifacts are interferences created by the human body [17]. These groups will be described in more detail in the next two sections.

Physiological Artifacts

As mentioned, Physiological artifacts are characteristics of the human body, including ocular movements, head movements, heartbeats, and muscle noises. For most of these artifacts, it is beneficial to record their corresponding biosignals to be able to retrace them in the measured signal [32].

Ocular artifacts can be measured by Electrooculography (EOG) and are mainly produced by eye movements and blinking. The latter generally produces a much larger amplitude than the EEG activity. Because of the anatomy of a human, EOG contaminations mostly occur in the frontal electrodes [32].

An even more common artifact comes in muscular activity, measureable by an EMG signal. Appearing in studies where the person is awake, the main actors for these kinds of contaminations are swallowing, talking, and walking, depending on the experimental environment. Most cranial EMG signals are in charge of disadvantageous effects in the recorded signal. However, it is difficult to detect and correct them as EMG not only represents a wide range on the spectral distribution, but also does not repeat, making it even more difficult to classify them [32].

Last but not least, the activity of the heart (ECG) plays a significant role in contamination of measured EEG signals. In contrast to EMG data, a regular pattern is visible, which can be further classified by comparing it to ECG measurements. This way, it is relatively easy to remove those from the recorded signal. [32]

Non-Physiological Artifacts

On the other hand, the so-called non-physiological artifacts are coming from outside-world interference. The most common source is electrical interference from power lines (50 Hz in Europe). This generally can be reduced by shielding the cable or moving the subject far away from the power source [17].

Another very common non-physiological artifact is the poor placement of electrodes on the scalp as the created DC potential is disturbed. Therefore, electrodes should be checked regularly to ensure high quality recordings [17].

Another obvious source for those kind of artifacts is the malfunction of any part within the measurement system. For example, amplifiers can create noise caused by the thermal agitation of electrons within the circuit since their are small and have a small single chip [17].

Lastly, digitial artifacts are artifacts originating from digitising, storing, or processing the EEG signal in a computerized system. The main reason for that is loose wiring, leading to mistakes in the analog to digital conversion and skew errors [17].

2.4.2 Artifact Removal Techniques

Before presenting state-of-the-art techniques for removing artifacts, it is essential to mention that most of these algorithms and techniques concern physiological artifacts since controlling the recording environment can correct or altogether avoid most non-physiological artifacts. By checking the placement of the patient and the electrodes

on the scalp, non-physiological artifacts remain a small problem in artifact removal compared to handling physiological artifacts [17, 32].

The first technique is the so-called (Frequency) filtering. It not only removes the power line interference but also witdraws high- and low-frequency noises. Depending on the goal of the filtering, different techniques are available, mainly a low-pass filter, a high-pass filter, a band-pass filter, and a band-stop filter. While a low-pass filter removes all frequencies above a chosen threshold, a high-pass filter eliminates frequencies below a boundary. The implementation of band-pass filters requires the definition of a lower and upper frequency boundary. When applying, it removes all frequencies below the lower and above the higher thresholds (so keeping all frequencies within the given constraints). Last but not least, a band-stop filter does the opposite of a band-pass filter. It removes all frequencies within the chosen interval, thus keeping the frequencies below the lower and above the upper bounds. Frequency filtering should be the first step of preprocessing EEG signals (even before segmentation) to remove low- and high-frequency noise outside the spectrum of the defined brain rhythms [17].

When discussing algorithms and techniques dedicated to removing artifacts, assumptions about the recorded signal must be met. The most important assumption is that the recorded signal is the sum of the brain activity and noise. A more generalised version describes the measured EEG signals as linear mixtures of waveforms descending from multiple brain sources and artifacts, ultimately propagating to the scalp. Mathematically, this can be described by the following mathematical equation:

$$\mathbf{X} = \mathbf{AS} + \mathbf{V}, \tag{2.1}$$

where $\mathbf{X} \in \mathbb{R}^{n \times N}$, $\mathbf{A} \in \mathbb{R}^{n \times m}$, $\mathbf{S} \in \mathbb{R}^{m \times N}$,$\mathbf{V} \in \mathbb{R}^{n \times N}$. While N represents the total count of samples measured, n stands for the number of EEG channels, and m for the number of total sources the signal originates from. In Equation 2.1, $\mathbf{X}$ describes the EEG signals resulting from the measurement. While $\mathbf{V}$ stands for added noise[3], $\mathbf{A}$ is an unknown mixing matrix and $\mathbf{S}$ is a matrix of unknown sources representing brain sources and artifacts [32].

With this model, the goal is now to solve the Blind Source Seperation (BSS) problem, so in this case, separating neural activity from artifact activity. For this problem, component-based methods are used because they find principal or independent components equal to the input EEG channels. After processing in the transformed domain, an inverse conversion is applied with the corrected components, resulting in processed EEG data minimising general noise and maximising the overall neural information in the signal [32].

In general, BSS aims to estimate $\mathbf{S}$ based on $\mathbf{X}$ in Equation 2.1. It must be mentioned that those methods' effectiveness lies in different assumptions that have to be met

[3] In the following, it is assumed that $\mathbf{V} = 0$

to a certain extent, e .g., independence or non-Gaussianity. The more fulfilled these assumptions are, the more effective the BSS method is. Apart from that, the success of such techniques also depends on correctly identifying the sources as either brain activity or artifacts, which can serve as a challenge depending on the extent of artifacts [32].

One of those methods is Principal Component Analysis (PCA), which uses an orthogonal transformation to turn the observed data to linearly uncorrelated variables (Principal components). The resulting components have the property of having the most significant possible variances while being orthogonal to each other. However, current research suggests that artifacts and cerebral activity can be modeled better as independent instead of orthogonal. To use this method, Independent Component Analysis (ICA) has emerged as the leading method for BSS [32].

ICA consists of multiple related methods, separating linearly mixed signals by taking advantage of the statistical independence of the sources. The algorithm usually works based on Higher-order statistics (HOS). After prewhitening the covariance matrix of the data, ICA tries to find a linear transformation $\mathbf{W}$ for estimated sources $\hat{\mathbf{S}} = \mathbf{WX}$. $\hat{\mathbf{S}}$ is then supposed to be as independent as possible. How effective the ICA is depends on the independence of sources and the full column rank of the mixing matrix $\mathbf{A}$. Since this algorithm relies on statistical features, the number of samples must be sufficiently large. In most cases, it is recommended to use all available data because cerebral activity and artifacts are spatially stationary through time [32].

2.5 Feature Extraction of EEG Signals

Before classification tasks can be performed on the denoised signal, it is essential to map the characteristics of each signal to specific metrics, called features. In biomedical signal analysis, big data has become more common, thus leading to high-dimensional sample data. Therefore, the biggest goal of feature extraction is to reduce this dimensionality and represent the data more compactly. Overall, feature extraction techniques not only allow for the reduction of complexity but also for saving on resources and computational time [29].

Different ways of extracting features from the EEG signal exist. Depending on the task and circumstances, distinct measures can lead to independent and different results. Therefore, it is important to explore as many different features across multiple domains as possible. Mainly, there are three domains from which features can be derived from, namely the time domain, frequency domain, and time-frequency domain, each extracting features based on different attributes of the EEG signal, capturing versatile characteristics of each EEG channel. For this reason, it is vital to further perform Feature

selection algorithms like the Minimum Redundancy Maximum Relevance (MRMR) selection technique to keep the dimensionality low and work with an optimal set of features. The following sections explain characteristics of the domain's features alongside examples, which is then concluded by presenting the MRMR feature selection method.

2.5.1 Time Domain Feature Extraction

In time domain feature extraction, signals are analysed to quantify how they change over time. Depending on the number of data points in a session, windowing may be employed to improve results [29].

One of the simplest and common approaches is calculating basic statistical features from the signal at hand. This includes mean, standard deviation, maximum, minimum, median, variance, range, skewness, and kurtosis [29].

In addition to these basic statistical features, parameters tailored for signals can be used to achieve desirable results and information. One of those is Root Mean Square (RMS), which measures the average power of a signal, representing a signal's energy or loudness. It can be calculated with Equation 2.2, where N is the number of samples and $V(i)$ the amplitude of the signal of the i-th sample [26].

$$V_{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} V^2(i)} \tag{2.2}$$

To depict the noisiness of the EEG signal, the Zero Crossing Rate (ZCR) can be measured. As the name suggests, it counts the number of sign changes and normalises the value with the number of samples.

In information theory, the (Shannon) entropy describes a signal's uncertainty and information content in bits. A low entropy means the signal is predictable, tending to a constant signal, whereas a high entropy represents a trend towards random noise [4]. The Shannon Entropy can be computed using Equation 2.3. p_i hereby stands for the probability of the i-th data point within the sample window [13].

$$H = -\sum_{i=1}^{n} p_i \log_2(p_i) \tag{2.3}$$

Additionally, used in recent EEG studies, the so-called Hjorth parameters gained more popularity as they are able to quantify power, frequency content, and signal shape variation. These parameters are three features, the activity (Equation 2.4), mobility (Equation 2.5) and complexity (Equation 2.6) of a signal $x(t)$.

$$h_0(x(t)) = var(x(t)) \tag{2.4}$$

$$h_1(x(t)) = \sqrt{\frac{h_0(\frac{dx(t)}{dt})}{h_0(x(t))}} \tag{2.5}$$

$$h_2 x(t) = \frac{h_1(\frac{dx(t)}{dt})}{h_1(x(t))} \tag{2.6}$$

Mathematically, each parameter stands for a specific characteristic of the signal. While the activity stands for the signal power, the mobility depicts the mean frequency and the complexity of the change in frequency [33].

2.5.2 Frequency Domain Feature Extraction

The frequency domain is an analytical space in which the EEG signal is interpreted in terms of frequency [8]. While in the time domain, the data is conveyed as a signal amplitude change, the frequency domain represents the change of its signal power. To do that, the signal has to be transformed from the time to the frequency domain by spectral estimation. This technique approximates the Power Spectral Density (PSD) of a function to detect periodicities in the data, leading to the decomposition of the signal into distinct sinusoids, emphasised in Figure 2.3. In this example, the signal can be described as three sinusoids with frequencies of 2 Hz, 8 Hz, and 20 Hz and amplitudes of 1, 0.8, and 0.5, respectively [17].

One of the most accurate methods to compute PSD is Welch's method based on a Discrete Fourier Transform (DFT). In general, DFT creates the representation of the signal in the frequency domain. A Fourier transform characterizes a time-dependent signal as the summation of multiple sines and cosines. Welch's method now divides the signal into K windowed data segments of M samples (overlapping by D samples) and computes the DFT over each segment. The average of all results then leads to the final spectral estimation. This approach causes a reduction in variance by $1/K$, making it more precise than other standard methods [17].

Multiple different features can be extracted from the PSD to access the frequency-related characteristics of an EEG signal. One of the most prominent EEG spectral features is hereby the band-limited power. It represents the spectral power within a particular frequency band, i. e., within the frequency range of each brain rhythm. It can be calculated by making use of Equation 2.7, ultimately being the sum of all power values P at each frequency point f_i within a frequency band I_{band}. For normalisation and elimination of individual differences, the relative power can be taken by dividing by the total power in the whole frequency range [17]. This method allows a detailed analysis of the different mental states during a session.

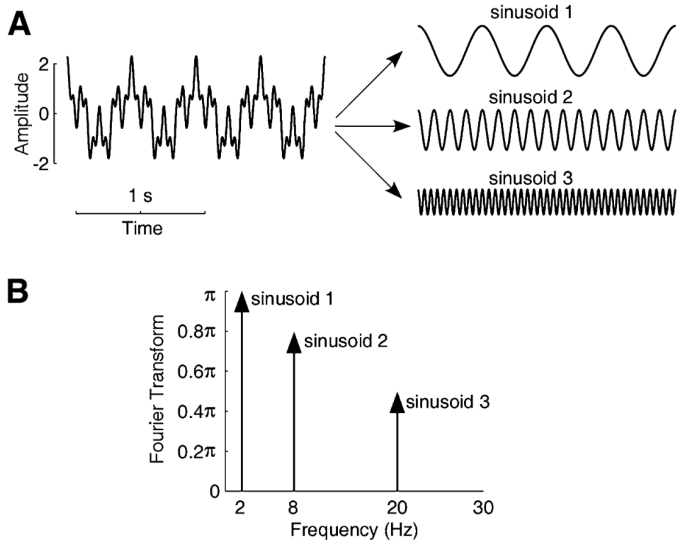


Figure 2.3: Comparision between Time and Frequency domain. From the signal (on the right), sinusoids (on the left) are extracted that have unique frequencies and amplitudes (at the bottom) [17]

$$P_{band} = \frac{1}{N} \sum_{f_i \in I_{band}} P(f_i) \tag{2.7}$$

Furthermore, the spectrum can be treated as a random process, thus calculating statistical features similar to those mentioned in the time domain[4] is possible [17].

2.5.3 Time-Frequency Domain Feature Extraction

Many useful features can be computed from the frequency domain. However, it is assumed that the spectrum of an EEG signal does not change over time. This premise oversimplifies the signal at hand since EEG is defined by non-stationary and dynamic behaviours, leading to a change of the spectrum over time [17].

This problem can be solved by combining the time domain with the frequency domain; Time-frequency Analysis (TFA) methods emerged to simultaneously analyse time and frequency characteristics. There are two main types of TFA techniques. The first one comprises methods that assign signal power per frequency point. These so-called time-frequency power distributions can be computed by a Short Time Fourier

[4] cf. Subsection 2.5.1

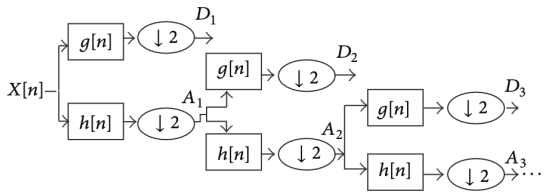


Figure 2.4: Visualization of a DWT [10]

Transform (STFT) or a Continous Wavelet Transform (CWT). The second class of techniques is called time-frequency signal decomposition. As the name suggests, it decomposes the signal into additive components concerning their Time-frequency properties. In that case, methods like Discrete Wavelet Transform (DWT) or matching pursuit can be employed [17]. In the following, Wavelet Transform techniques will be explained.

Generally, wavelet transform methods compress a time-varying signal into a few parameters representing the signal, called wavelets. At the core, wavelets are localised, small wave-like functions that refer to the different positions and scales of the signal. Based on that, a wavelet deocmposition can analyse various data characteristics depending on the window size. While long windows extract low-frequency components, short windows get high-frequency information. This windowing technique enables capturing not only coarse but also fine details [10].

A mother wavelet function ψ is used to generate the single wavelets by shifting and scaling ψ. To achieve this, two parameters play a key role. Firstly, the scale parameter a is responsible for determining the wavelet's resolution. Secondly, the translation factor b controls the shift of the time domain. In general, there are two types of Wavelet transform methods: the CWT and the DWT.

To compute the CWT coefficients of a signal $x(t)$, following equation is used:

$$CWT(a,b) = \int x(t)\psi^*_{a,b}(t)dt \tag{2.8}$$

The signal is probed over every defined combination of a and b to calculate a 2D-matrix, where each entry corresponds to a specific coefficient [10].

One of the biggest problems of CWT is the result of a lot of unused information and computational effort due to the continuous changes in a and b. To solve this issue, DWT can be employed instead, which is showcased in Figure 2.4. Each step (or level) consists of two digital filters $g[n]$ and $h[n]$ and two downsamplers by two. $g[n]$ stands for the mother wavelet function and acts as a high-pass filter, while $h[n]$ is its mirror function working as a low-pass filter. Each stage outputs a detail of the signal D and

an estimation of the signal A, which is then used as an input for the next step. This results in a hierarchical set of coefficients, which become more accurate the higher the level is set [10].

2.5.4 Feature Selection

Based on the previous sections, it becomes clear that many different kinds of features can be calculated based on an EEG signal. The number of features, especially scales, in EEG based classification problems, since all features are computed for each channel and the metrics are very versatile. However, depending on the dataset, many features can lead to undesirable results, such as Over- or Underfitting. Therefore, it is vital to employ standard feature selection methods that choose the optimal subset of all these metrics. This not only leads to a reduction in dimensionality but also to choosing the parameters that hold the most information. One feature selection method in that domain is the MRMR feature selection, which will be explained in the following.

As the name suggests, the MRMR algorithm tries to find the feature set that has maximal relevance and minimal redundancy between each other. On the one hand, relevance V describes how much information a feature set F provides about the different class labels. On the other hand, redundancy W defines how much information is shared between features in the feature set [18].

With both W and V at hand, the goal of the MRMR algorithm is to maximise W while minimising V. The mutual information quotient $MIQ = \frac{V}{W}$ or the mutual information difference $MID = V - W$ can be considered for this. To find the optimal feature subset S with size M, features $f \in F$ are iteratively added to S to maximize the criterion. However, this method can be computationally expensive since it has an exponential runtime depending on $|F|$ and M [18].

3 Related Work

3.1 Music Genre Classification with Audio Features

Before discussing how music genres can be derived from EEG signals, it is important to briefly mention research concerning Music Genre Recognition (MGR) based on the song itself. This serves as proof of concept that audio features differ between different genres. To do that, recent research is presented in this section, focusing on features and classic Machine Learning (ML) models used.

Many different datasets can be used to perform MGR, which are also highlighted in Chapter 4. Such datasets are mostly well-known and are already labelled based on their genre, like the GTZAN dataset used in studies by Tzanetakis et al. and Haggblade [31]. Other datasets that require some labelling beforehand also play a vital role in the field of MGR because some music clips are not clearly annotated regarding the music genre. Such datasets are, for instance, the Audio Set [3], the Million Song Dataset [23], and the TagATune set [27]. Although many different datasets have been used in past research, many genres overlap. For instance, Pop, Rock, Hip Hop, Electronic, and Classical are in the mentioned datasets. Additionally, genres like Jazz, Metal, Reggae, and Dance appear multiple times across the datasets brought up [3, 23, 27, 31]. Related to this, the number of genres also differs depending on the study and the dataset used. While Tznaetakis and Liang perform their research on ten different genres [31, 23], Rotheneder and Bahuleyan use 16 and seven genres respectively [3, 27]. Besides that, Haggblade et al. filtered the GTZAN set according to the four most distinct genres: Metal, Pop, Jazz, and Classical [15].

Features from the corresponding audio signals are computed to reliably represent the music clips. Feature extraction methods in this research area are versatile, leading to different approaches from study to study, which also depend on the classification models used. In the domain of Deep Learning, Convolutional Neural Network (CNN)s were often used in this area. As they excel in image recognition and classification, the spectogram[1] is taken as input from which the CNN builds optimal feature maps [3, 27]. However, classifiers in the ML domain still require hand-crafted features derived from the audio signal. The most prominent features come from the frequency domain

[1]Visual representation that shows how the frequencies in a signal change over time, with time on one axis, frequency on the other, and color/intensity indicating the strength of each frequency.

and specifically revolve around the Mel Frequency Cepstral Coefficients (MFCC)[2], also considered as timbre. Accompanied by the timbre, other frequency-dependent features include spectral centroids, bandwidth, and rolloff [3, 15, 23, 31]. Apart from the frequency domain, Bahuleyan and Liang also computed time-domain features, incorporating central moments, ZCR, tempo, and loudness [3, 23]. Besides that, unconventional features like rhythmic content [31] and lyrics features [23] are considered to improve the information held by those parameters.

To finally predict musical genres based on audio signals, different statistical models have emerged to find patterns in data to reliably classify the music clips. Similar to the feature extraction, these models can be differentiated between Deep Learning (DL) and ML methods. The former is mainly represented by CNNs used by Bahuleyan and Rotheneder. They achieved an accuracy of 64% and 77.3% respectively [3, 27]. Classic Neural Networks have also been used in a recent study by Haggblade et al., achieving 96% on their pre-selected four genres [15]. Although CNNs seem to yield good accuracy results for MGR, classic ML classifiers are still employed frequently. Most notably, Bahuleyan also analysed the performance of feature engineering-based models, namely Logistic Regression (LR), Random Forest, Extreme Gradient Boosting (XGB), and an Support Vector Machine (SVM). Interestingly, none of these were able to outperform the CNN as the highest achieved accuracy was 59% by the XGB. This gives a hint that DL models (especially CNNs) can perform better results than classic approaches [3]. Supporting this statement, Haggblade et al. also compared the Neural Network's performance with other ML methods, including SVM, k-Means, and K-neares neighbor (K-NN), where none of these were able to reach a higher accuracy than the Neural Network. Apart from that, classic statistical models like K-NN and Gaussian Mixture Model (GMM) also see usage in audio-related MGR. On the ten genres, an accuracy of 61% was achieved by the GMM and 60% by the K-NN [31]. From the perspective of the thesis and the goal of classifying music genres based on EEG signals, the fact that ML techniques can identify differences across genres supports the fact that the neural response should be different from genre to genre, which can be captured in the EEG data. These differences and patterns can then be identified by the ML model in theory.

3.2 EEG Datasets and Studies with Music Stimulus

As music is a popular stimulus used in EEG studies to collect the signal data, many research projects have been conducted in this area. This section intends to serve as an overview of work that has been done in the context of constructing an EEG

[2]Features extracted from audio signals that represent the shape of the spectral envelope

dataset where music served as a stimulus. For each dataset, the experiment setup and acquisition methods are described, and certain preprocessing and denoising steps are emphasised.

3.2.1 DEAP

The Database for Emotion Analysis using Physiological Signals (DEAP) is an EEG dataset made public in 2012 by Koelstra et al. It is a multimodal dataset providing data for analysing human affective states. For 32 participants, EEG and peripheral physiological signals were recorded while watching 40 one-minute sections of music videos. The video was also shown alongside the music, serving as an additional stimulus. After each snippet, test subjects were asked to rate each music video in terms of valence (positively or negatively connoted), arousal, liking, dominance and familiarity [19].

The music comes from a website called Last.fm and was selected using a sophisticated procedure. 120 music videos were initially selected and labelled with keywords out of a list of emotional keywords provided by the Last.fm website. Based on each emotional keyword, ten songs labelled most with one of them have been selected in the next step of the procedure. The chosen songs were further filtered according to valence and arousal attributes. After this filtering step, a one-minute highlight section was selected with an effective highlighting algorithm from each music video. Last, a web-based assessment experiment was conducted to decide the 40 final music video clips for the DEAP study [19].

The experiment was conducted in a laboratory environment with controlled illumination. The EEG signals have been recorded with a sampling rate of 512Hz by 32 active AgCl electrodes placed in the international 10-20 system. Apart from that, not only have physiological signals been measured, but frontal face videos for 22 of the 32 participants have also been recorded. The basic procedure of each session includes a five-second baseline recording before displaying the one-minute music video. When the video ended, the participant assessed the clip based on arousal, valence, liking, familiarity and dominance[19].

For further investigation of the recorded signal data, the researchers provide a preprocessed variant of the EEG data. First and foremost, the data were commonly referenced and then downsampled to 256Hz. The downsampled signal went through a high-pass filter with a 2 Hz cutoff frequency employing the EEGlab toolbox. Eye artefacts were removed by using a BSS technique. For further analysis, the baseline recording of five seconds was removed. [19]

3.2.2 Music BCI

Behavioural responses like in the DEAP dataset can be researched, as well as the perception of musical patterns. Therefore, researchers from Berlin and Copenhagen have chosen polyphonic music as a stimulus for an EEG dataset. Polyphonic music describes music with two or more parallel melodic lines, including different instruments. In their work, the main goal is to explore the suitability of polyphonic music in Brain-Computer Interfacing (BCI) [30].

For this purpose, eleven participants aged 21-50 years were chosen, each not educated in the field of BCI research. For acquisition, a 1000 Hz sampling rate was determined with 64 electrodes placed according to the international 10-10 system. The stimuli were 40-second music clips, each of which contained six tracks. Two of these tracks are meant for stereo audio playback, the other four are a mono mix of the stereo clip recorded as additional EEG channels. Each clip contains three instruments playing musical patterns, where one instrument plays a deviant version while the other two stay the same. Per instrument, there are 3-7 deviations per clip. Synth-pop, which has around 130 bpm, and jazz, which has around 120 bpm, were chosen as genres. While the Synth-Pop clips consist of drums, a bass, and a keyboard, the clips coming from Jazz comprise flute, piano, and a double bass. Ten different clips were made for both music types with those kinds of deviant patterns. [30]

During the experiment, the participant was seated comfortably in a chair and instructed to stay still and relaxed, especially to minimise eye movements during the trial. The main experiment is divided into ten blocks with 21 music clips played randomly. The listener was told to pay attention to each of the three instruments in seven different music clips. To further ensure attention, the participant has the task of counting the number of deviants in the cued instrument. Before the following clip, the participant must enter the recognised number of deviations [30].

The signal data were preprocessed to further analyse the recorded EEG data. After downsampling them to 250 Hz, they were low-pass filtered using a Chebyshev filter accompanied by a passband of 42 Hz and a stopband of 49 Hz. The data is sectioned into epochs from -200 ms to 1200 ms for additional denoising. While the prestimulus interval was used for baseline correction, a min-max criterion was employed to discard physiological artefacts [30].

3.2.3 MUSIN-G

In 2022, researchers published an open-source EEG dataset examining the correlation between the brain and music genres. In the experiment, 20 participants listened to naturalistic songs (meaning mostly without lyrics) across 12 genres, ranging from

Indian folk music to Goth Rock or Western electronic music. Apart from the genre, subjective characteristics like familiarity and liking have been considered [25].

It is essential to mention that all participants have little musical training and are therefore considered non-musicians. The participant's EEG data were recorded by a 128-channel high-density EEG cap while sitting comfortably in a dark environment. The main procedure can be divided into two parts. At the start of the acquisition, a baseline recording was conducted during absolute silence lasting for two minutes. During this process, the subject's eyes were closed. The second part then consists of listening to the songs, where the order is randomised depending on the person hearing the audio. A beep sound instructed the participants to close their eyes when the song started to play. After each song, a ten-second gap was chosen, followed by another beep sound indicating to open their eyes for answering two questions. The participant has to rate the song on a scale of one to five regarding familiarity and liking. After answering the questions, the next song was played with a trailing ten-second gap in between [25].

Since the data is provided as raw and segmented EEG data, the signals can be preprocessed for further analysis. The authors suggest a workflow that includes the EEGLAB toolbox in MATLAB. First, the signals are high-pass linear FIR filtered at 0.2 Hz. This is followed by the CleanLine method, which removes 50Hz of line noise (comparable to a notch filter). Before segmenting the data into each song, the filtered data were down-sampled to 250Hz. By employing a spectrum criterion, bad channels were removed to prepare them for the artefact removal. The Multiple Artifact Rejection Algorithm (MARA) was used to remove the artefacts, an ML-based method for artefact detection and removal. Lastly, removed channels are interpolated with the spherical function of EEGLAB before the average reference is applied to the whole dataset [25].

3.3 Music Genre Recognition with EEG Signals

With the help of EEG datasets similar to the ones described above, researchers can use the acquired data to perform classification tasks in affective computing. Using personal data like brain signals can capture content-centric features like personal preference, which would not be possible with audio features [12]. Most of this research done in this field deals with predicting a person's emotional state, which will be covered in Section 3.4. Although not many studies regarding MGR based on EEG signals have been published, a recent study is presented in the following.

Ghamemaghami and Sebe studied whether performing MGR based on brain-recorded signals is possible. As a source of EEG data, the authors employed the preprocessed version of the DEAP dataset[3]. Because the dataset does not provide labels for genres,

[3] cf. Subsection 3.2.1

all 40 song clips were manually annotated in two distinct categories, POP and ROCK. While POP comprises Pop, Dance, Disco and Techno, ROCK consists of Rock and Metal [12].

As features, the averaged power over each of the four brain rhythms (alpha, beta, theta, gamma) was taken into account. Furthermore, multimedia content features were calculated from the song's audio and music video as a comparison. With the features at hand, an SVM classifier was used with a leave-on-clip-out-cross validation technique, measuring accuracy and F-score. It was classified not only on the level of subjects (i. e., only using the brain data from each subject) but also on the whole population (i. e., taking all subjects into account). On the subject-level, the researchers achieved a better accuracy than chance level when only employing EEG signals, with a variance of 10% around an average of 60%. When fusing the EEG signal with the multimedia features, the classifier provides an even higher accuracy compared to only using EEG signals, leading to an higher average of accuracy of 75% and a variance of 5%. The F1-scores are around the same range respectively [12].

For population-level analysis, the subject-level descriptors were considered to calculate the genre over a majority-vote, outperforming both accuracies of subject-level evaluation [12].

This research shows first evidence that brain signals like EEG can be used for classifying music genres, demonstrating the potential for user-centric music retrieval. Furthermore, since the fusion of brain signals with multimedia content has shown significantly better results, it suggests that there is complementary information between both [12].

3.4 Emotion Recognition with EEG signals

Since work in EEG based MGR is limited, it is vital to investigate further research on classification tasks related to EEG signals. As listening to music is highly linked with the emotional state of the human brain, it is possible to extract patterns out of a measured EEG signal to map them to different emotional parameters like valence or arousal. As those reactions are highly correlated with the music's genre, it is important to get an overview of recent studies regarding Emotion recognition using neural activity. Before starting with the review, it must be noted that the main focus lies on studies using classic ML methods as this thesis aims to only make use of such methods as well.

As a basis, it is important to clarify where the EEG data orignates from that is used for classification. This is essential since the results and performance of classification tasks heavily depends on the EEG signal. To ensure reproducibility and comparision between different work, most studies employed publicly available EEG datasets that are

specifically designed for emotion tasks. Very prominentely are hereby the DEAP[4] [2, 6, 16, 22], the SJTU Emotion EEG Dataset (SEED) [16, 22] and the Muse Headband EEG dataset [1, 7]. While the former one uses music as stimulus, both latter ones induced emotional states by showing clips from different movies. Besides well-known datasets, Lin et al. performed their own experiment involving simple music listenting to fully control the quality of the brain signal [24].

Linked to the source of EEG signals, it is essential to clarify what exactly will be predicted. Most of the studies use the two-dimensional valence and arousal model. It maps both valence (how positively/negatively connoted is the emotion) and arousal (how strong/weak is the emotion) on a coordinate system to map them to emotions. This leads to much versatility in emotion mapping. The most common one includes prediciting either valence or arousal seperately. To make the task easier, a threshold is chosen which can divide valence/arousal in multiple categories. For example, a binary classification for high/low valence/arousal values [1, 16, 22] or even ternary classification with high, low and neutral as labels [7]. Besides that, valence and arousal can be composited together to map the values to real emotions. As a reference, Lin et al. defined four emotions (joy, anger, sadness and pleasure), where each of which is occupied by one quadrant of the coordinate system [2, 24]. Overall, instead of using discrete emotions, continous models like the valence-arousal model are employed to classifiy emotional states.

Besides the EEG signal itself, feature engineering is crucial for the performance and accuracy of classification. As EEG signals are very versatile, many different features can be extracted from the neural activity. In various related research, parameters have been extracted from the time-domain capturing time-dependent attributes of the signal. Besides classic statistical features like mean, standard deviation, etc. [7, 34], the most used temporal features included (differential) Entropy and Higuchi's Fractal dimensions [2, 16].

Moreover, most of the studies that perform classification on EEG data use frequency related features as it is possible to extract information from certain frequency bands. In this case, these frequency bands can be defined for each brain rhythm, hence being one of the most precise measures for brain activity and is therefore commonly used in EEG based classification tasks [1, 22, 24]. This band powers, however, can still be further processed by considering the spatial attributes of the EEG channels and the anatomy of the brain. By computing differences, quotients, etc. between two opposite electrodes can give valuable information on the emotional state of person [24].

To combine both the time and frequency domain, wavelets are also utlised to add a time component to the popular band power features [34].

[4] cf. Subsection 3.2.1

Different ML approaches were used across the mentioned studies to recognize emotions. The most used models include, SVMs [1, 16, 22, 24, 34], Random Forests [1, 7] and XGBs [1, 7, 16]. Apart from the more popular models, studies also have applied K-NN [2, 16], naïve Bayes [2], Decision Trees **alhalaseh2020machine!** (**alhalaseh2020machine!**) and LR [1]. However, not only have single standalone models been trained, but also stacking ensemble classifiers. On the one hand, the base classifiers consist of a Random Forest, a Light Gradient Boosting Machine, and a Gradient Boosting Classifier. On the other hand, a second Random Forest model is trained on the base classifier's output and is used as the meta-classifier [7]. Moreover, DL methods have also found utilisation in the form of CNNs [2].

In conclusion, many different studies demonstrate that emotion classes can be reliably predicted based on the valence and arousal model. This is an essential conclusion in this thesis since musical perception is highly correlated with someone's emotional state.

4 Music Datasets

Since this thesis deals with music genre classification, a suitable set of musical stimuli has to be chosen to gather valuable data. As research in the domain of music is widespread, multiple music datasets exist that are used in different studies. Therefore, the difficulty lies in choosing the right dataset for this thesis. However, the work goal crystallises precise requirements for such a music dataset.

First of all, the songs that are listened to are supposed to be around 30 seconds long. The reason is that the stimulus could be lost if the song runs too long because the brain can adapt to the genre-specific features, thus reducing the visibility of the impact on the brain. Apart from that, the number of songs should be large, with around 800 to 1000 song clips. A large pool of songs is essential to get as much different and versatile data as possible. This helps in future steps regarding classification. However, the genres used in this pool must also meet specifications related to this requirement. First, the songs must be clearly labelled to avoid confusion in later stages of the study. It is also essential to know how balanced the dataset is in terms of genres listened to. This is supposed to be as balanced as possible so that each genre is represented equally. Additionally, it should be easy to employ this dataset in the study. It should be uncomplicated to load the songs and get access to labels or other important information.

In the following, different popular and public music datasets are presented. They will be analysed according to their possible use in this research and how they meet the requirements specified above. After that, the reasons for the final choice of dataset are outlined.

4.1 Audio Set

In 2017, Gemmeke et al. published a dataset for audio events called Audio Set[1]. This is a large-scale dataset of audio events labelled by humans using a carefully structured hierarchical ontology of 632 audio classes. The source of the audio events is derived from 10-second segments of YouTube videos. These audio events are occurrences in the real world, like human sounds, animal sounds, natural sounds, sounds of things, source-ambiguous sounds, environment sounds and music. Then, each sound

[1] `https://research.google.com/audioset/`

is labelled with one ontology and a maximal hierarchical depth of six. In determining labels, the researchers ensured that the tags were intuitive and clearly distinct. [11]

For this project, it is a big advantage that all songs are publicly available in form of a YouTube link. For that reason, however, seammless integration is harder to accomplish because the individual songs have to be downloaded first. Another drawback is, that this dataset's focus is not on music but on general audio events. This is why, in the first step, the dataset would have to be filtered to contain only the music clips. However, the fact that the clips are only 10 seconds long means there is a risk that this is not enough stimulus to arouse the brain and see the distinct patterns. A further problem lies in the format of the data. Since the audio event labels range from many different occasions, only the audio events under the category of Music are important in this case. From this subset, only the songs with one of seven genre labels are required: pop, rock, hip hop, techno, rhythm and blues, vocal, and reggae. In total, there would be 40540 clips available. However, the songs are not balanced in regards of their genre. [3]

4.2 Million Song Dataset

The Million Song Dataset[2] is a collection of audio features and metadata for a million music tracks, which is also freely available. This is the largest research dataset in the field as it contains 1,000,000 songs, accumulating around 280GB of data. In addition to the songs, mainly from western commercial music in 2011, general metadata and audio features like pitch, timbre, etc., are provided using the EchoNest API. This API is an online resource supporting many music applications on the web. Since most of the "natively" available metadata comes from the EchoNest API, it is easily extendable to further labels like genres. [5]

Although this dataset may seem like a great fit because of its rich amount of songs, the number of songs is too much, since only 800-1000 songs are used during the acquisition, most of the dataset would have to be filtered out. Furthermore, not all songs are trimmed down to a shorter clip but are provided in full length, which does not meet the requirement for this study. Moreover, the genres would have to be added externally since they are unavailable in the native dataset. This makes the process of genre balancing much more difficult than with datasets where the labels are already present within the original dataset.

[2] `http://millionsongdataset.com/`

4.3 MagnaTagATune

One challenge when creating such datasets is determining a label or tag for each song. In the context of genres, how is it decided which genre a song belongs to, as there could be overlapping features. Based on this hurdle, Law et al. created a music dataset to evaluate different tagging algorithms called MagnaTagATune[3]. It contains 25,863 music clips, each 29 seconds long. These excerpts were taken from 5223 songs. These music pieces range across genres like Rock, Pop, Jazz, Blues, World, Metal, Punk and many more. Each clip is generally accompanied by a vector of binary annotations of 188 tags using the TagATune game. [14]

TagATune is a game with two players that aims to collect music tags from the players. Each player is presented with a music clip, which they are assigned to label. After that, both players see what the other player has chosen as tags. Based on this information, the players must decide whether they listened to the same music clip. [21]

From the perspective of this study, this music dataset has the advantage that each clip is around 30 seconds long, hitting the desired length for one stimulus. Furthermore, it is beneficial that the songs vary in the genres used, thus making them a good candidate for a music dataset. However, there are some drawbacks, especially lying in the labelling of each clip. Although the labelling process is well-thought-out and transparent, each song can have more than one label. Depending on the players, one clip can be annotated with more than one genre tag. And since there are in total 37 genre-related tags [27], deciding which genre is the most suitable choice for a song is very difficult. Related to that, since there are around 26,000 clips, one would have to remove a majority of songs to let it fit into the scope of this thesis. Combined with the fact that multiple genres can be chosen for each song, achieving a balanced dataset in terms of genre is difficult.

4.4 FMA Dataset

The Free Music Archive (FMA)[4] is an open dataset designed for multiple tasks in the domain of Music Information Retrieval (MIR). It consists of 106,574 tracks distributed over 16,341 artists and 14,854 albums. Although this dataset offers many different audio features and metadata about the songs, it is most tailored to music genres. Each track is arranged in a hierarchical taxonomy of 161 genres. As a labelling technique, the researchers labelled the song according to the opinion of the artist who sings the respective song. Therefore, subjective errors in tagging and labelling could be

[3] `https://paperswithcode.com/dataset/magnatagatune`
[4] `https://github.com/mdeff/fma`

minimised. In addition to that, all tracks are not only stored in simple .mp3 files, but also are fully licensed. To make the use of the FMA dataset easier, different subsets of the complete dataset are offered depending on the research question and scope. In total, there are four subsets one can choose from. The first one is called *Full*, which has every track untrimmed and is therefore unbalanced in genre. The *Large* subset contains the same tracks as the *Full* set, but now, the audio is limited to 30 seconds each, extracted from the middle of the track. To tackle the simpler problem of single-label prediction in genres, the *Medium* subset only contains those songs that have only one top genre, resulting in 25,000 30-second audio clips. Furthermore, the number of genres is reduced from 161 to only 16. However, this subset remains unbalanced. Defferrard et al. have constructed the balanced *Small* subset to solve this challenge. It holds 8000 tracks, each being 30 seconds long. From all the genres, the eight most popular ones were chosen, and based on this choice, 1000 clips were chosen for each genre. The eight genres in this case are Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop and Rock. [9]

The FMA dataset brings many favourable characteristics for this research. The *Small* Subset of this set is a perfect fit since it is balanced according to genres and does not contain that many songs. However, around 900 songs per genre would need to be removed to have a reasonable number of songs for the scope of this thesis. Another advantage is the recency of the publication. It is frequently used in different areas of MIR and is accompanied by many code examples and setup guides.

But, despite of being balanced in regards of the genres, the genres are not 100% clear resulting in unclear behaviour and characteristics of the songs. For instance, the genres International and Instrumental are not comprehensive, since many genres can either have a cultural impact or don't have any lyrics. Another disadvantage is that Classical music is not included, since this type of music is well-known for impacting the brain.

4.5 GTZAN Genre classification

Last but not least, there is the GTZAN Genre Dataset[5], which is a dataset specifically designed for MGR. There are 1000 songs in total originating from 2001 and 2002. Each track is trimmed to 30 seconds and ranges over 10 different genres. These genres are widespread, including blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae and rock. Furthermore, each genre contains 100 songs, resulting in a balanced song set in terms of genres. All audio was gathered from different sources like CDs, radio, microphone recordings and more to vary in the conditions of the recordings. Apart from the audio files, the dataset also contains audio features for each song. [31]

[5] `https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification`

Dataset	Number of songs	Average duration	Genres	Balanced option
Audio Set [11]	40,540	10s	7	no
Million Song Dataset [5]	1,000,000	219s	undefined	no
MagnaTagATune [21]	25,863	29s	undefined	no
FMA Small [9]	8,000	30s	8	yes
GTZAN [31]	1,000	30s	10	yes

Table 4.1: Overview of potential Music datasets. The attributes only correspond to the parts of the dataset that are interesting in terms of MGR. The term undefined is used when there are no clear genre tags or if they depend on subjective interpretation and individual preprocessing

As this dataset is specialised in studies regarding MGR, it is an excellent candidate for the thesis's purposes. With its 1000 songs, it fits perfectly in the right range of songs. Moreover, since each genre consists of 100 songs, it is perfectly balanced, meeting another requirement for the experiment. Another advantage is that the genres are clearly described, from which clear attributes can be derived. The only drawback in choosing genres is that some genres overlap in characteristics. For instance, jazz originates from blues, thus sharing a lot of common characteristics. Nonetheless, many of the other genres have apparent differences from one another, like classical and metal.

4.6 Final Decision

All datasets presented in the previous sections have their strengths and drawbacks. An overview of the key attributes is shown in Table 4.1. Based on the evaluation of each dataset for the thesis's purposes, the most suitable song source for this work will be discussed and explained.

As mentioned, the audio set does not fit the requirements well. Since its focus is not on music but audio events, applying the set of clips in this use case is difficult. Moreover, ten seconds for each track is insufficient to arouse specific brain activity for each genre.

On the other end, the Million Song Dataset is not beneficial either. Although it only consists of music tracks, they are not trimmed and are full-length. Apart from that, 90% of all songs would have to be removed since only around 1000 tracks are necessary for this study, which can be a significant overhead (especially when ensuring a balanced set of songs).

From all the datasets, the MagnaTagATune dataset has the most sophisticated model

of tagging each clip, which tends to higher quality in labels. Another benefit is that all songs are trimmed to 29 seconds, meeting the duration requirement. However, this dataset is not focused on MGR and consists of 188 music-related tags. Moreover, the tags are binary vectors, so some songs could be multi-labeled, resulting in more difficult handling and unclear distinctions of songs and the genre they belong to. For this reason, the dataset was not chosen for this research.

Both the FMA and GTZAN dataset have very similar characteristics as they are specialised in MGR and therefore are balanced in genres, which is one of the most critical requirements in this study. Additionally, all clips are available as audio files with a constant duration of 30 seconds. The key difference lies in the size and genres present in the dataset. While the FMA set contains eight genres with 1000 songs each, the GTZAN consists of ten genres with 100 songs each. Besides, compared to the genres represented in the GTZAN dataset, the genres in the FMA collection are not very descriptive and (for some of them) difficult to distinguish. Therefore, the GTZAN song collection is chosen for the experiment setup and as a source of songs. Its easy file structure and naming convention can be seamlessly integrated into any setup, thus being the most suitable fit for this thesis.

5 Methodology

5.1 Acquisition

5.1.1 General setup, environment and stimuli

The experiment has been conducted in a laboratory setting. As mentioned in Chapter 4, the GTZAN Music dataset is chosen as a basis for the music stimuli. However, for the sake of the quality of the data, it would not be beneficial to hear all 1000 songs in one session. To avoid quick attention loss during a session and gather as much different data under different circumstances, the whole set of audio files is partitioned into 20 distinct sessions, each lasting around 30 minutes. This means that per session, 50 songs are played[1]. For each of these songs, not only are EEG signals measured, but also other biosignals like Photoplethysmogram (PPG), Inertial Measurement Unit (IMU) and temperature. However, this work will solely focus on the recorded EEG data. During each session, five songs from each genre are being played to ensure that each of the ten genres from the dataset is represented. This also ensures no genre bias in the data, thus resulting in a balanced signal across genres. During listening, the order of the songs is randomised to evoke a brain reaction that is as natural as possible, as the brain cannot predict the genre that is being listened to. Additionally, a five-second break is implemented in between songs to not overstimulate the brain.

All 20 sessions have been recorded over the course of two weeks, with an average of two sessions per day. Between each session, there was a short break of between ten and 30 minutes. This verifies a sufficient pause of the brain to fully prepare for the next session. The test person is the author, a 21-year-old, right-handed college student without any common (brain) diseases. To further guarantee a high quality in the acquired signal data, the experiment's environment and setting must be adapted. First and foremost, the test person is instructed to sit comfortably in a chair with his eyes closed to minimise artificats coming from blinking and eye-movement. Furthermore and most importantly, any movements, especially in the head region, must be limited and as minimal as possible since the EEG device reacts very sensitive to head movements. Apart from that, the room was quiet and isolated with dimmed natural lighting

[1] Because of a corrupt audio file, session ten only consists of 49 songs

combined with the brightness from a desktop monitor. The songs were played over the EEG headset (cf. Subsection 5.1.2) with average and constant volume.

5.1.2 Acquisition device

A non-traditional approach is chosen to record the EEG signal data coming from the brain. For this study, a prototype for a multimodal biosensing platform is chosen. It is a Bluetooth headset that can play music and measure biosignals at the same time. For connecting with a standard PC, it uses classic bluetooth and low-energy bluetooth for audio and biosignals respectively.

The headphones have multiple sensors and electrodes to measure biosignals like EEG, PPG, IMU and temperature. For EEG, the headphones are supplied with 2 × 4 sensor electrodes and two bias electrodes. The sensors are easily configurable and provide an ultra-low-noise front-end, meaning that the data points are amplified with minimal distortion and interference. Besides that, the electrodes sample with 24-bit and a frequency of 250 Hz.

For PPG, one optical sensor on the right side provides a high resolution of 19-bit ADC with a configurable sample rate. They also support raw data output or built-in algorithms to conclude heart rate or SpO2.

The IMU data is sensed by a 3-axis digital accelerometer fused with a 3-axis digital gyroscope. On a full scale, the accelerometer can recognize acceleration from $\pm 2\,g$ up to $\pm 16\,g$ (in steps of the power of two). The gyroscope can measure angular velocity from $\pm 125\,°/s$ up to $\pm 4,000\,°/s$. The recognised data is then combined to provide accurate information on orientation, movements and position.

Last but not least, there are three digital thermometers, one on the skin, another one inside the earcup and one on the outside of the earcup to measure ambient temperature. All of those temperature sensors have an accuracy of $\pm 0.1\,°\mathrm{C}$.

Since this headset is still a prototype, there still is ongoing development and research on the headset. Nonetheless, some small-scale research projects already employed the biosignals from the headphones. These projects range from stress detection using the PPG data to device control and the recognition of cognitive fatigue using the EEG sensors.

5.1.3 Technical setup

Multiple technical components have to come together to gather the experiment's data. All the components employ the Lab Streaming Layer (LSL). It provides a standardised way to stream time-sensitive data between devices and applications. Devices or apps

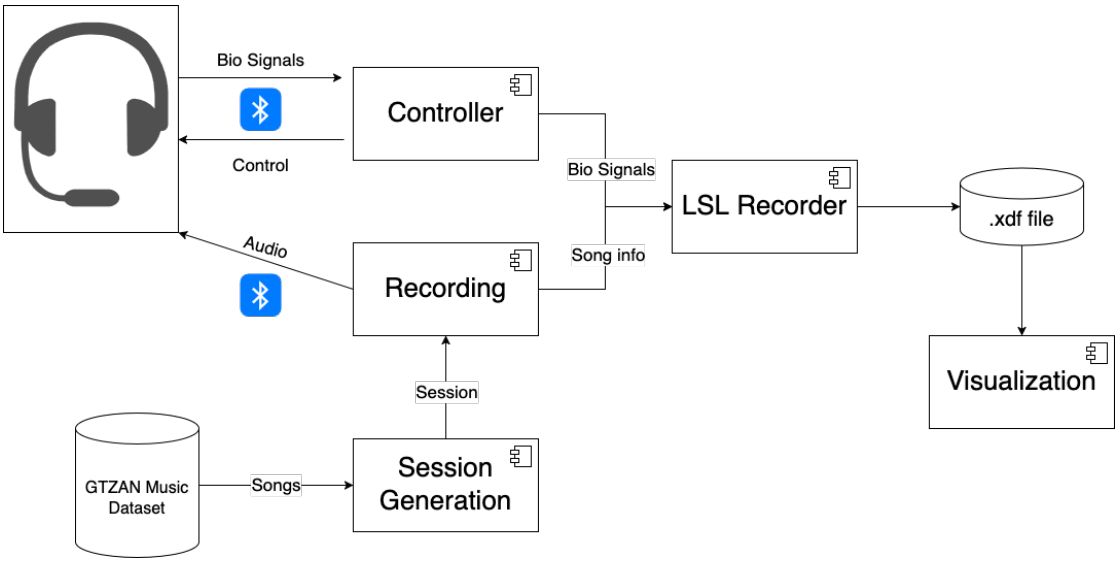


Figure 5.1: Technical setup

then push data on this stream, which is captured and stored in a special file form called .xdf. The workflow is illustrated in Figure 5.1 and will be explained in the following.

At the core, there are two main programs responsible for the data. On the one hand, there is the headset platform called acting as a Controller. This program is responsible for the sensor setup and starting and stopping the signal acquisition. Communicating with the headset's low-energy Bluetooth channel reliably sets up the biosensors of the multimodal headphones, making the device ready for acquisition. After a successful setup, the data is pushed on the corresponding LSL Stream. On the other hand, the study uses a custom Python script having two tasks regarding playing the music. Firstly, it loads all the songs from one of the 20 pre-generated sessions and randomly shuffles them. After that, the program plays each song one by one. Simultaneously, the script also pushes timestamps during recording to assign each segment to each song later.

By collecting all open streams on which the device/program push their data, the LSL recoder combines both components mentioned above. The collected information for each session is then stored in a seperated .xdf file, allowing for easy reading and visualisation.

5.2 Denoising and Preprocessing

After measuring and recording the EEG activity, the next step is to preprocess the acquired signal. Apart from segmenting the data into their respective song listened

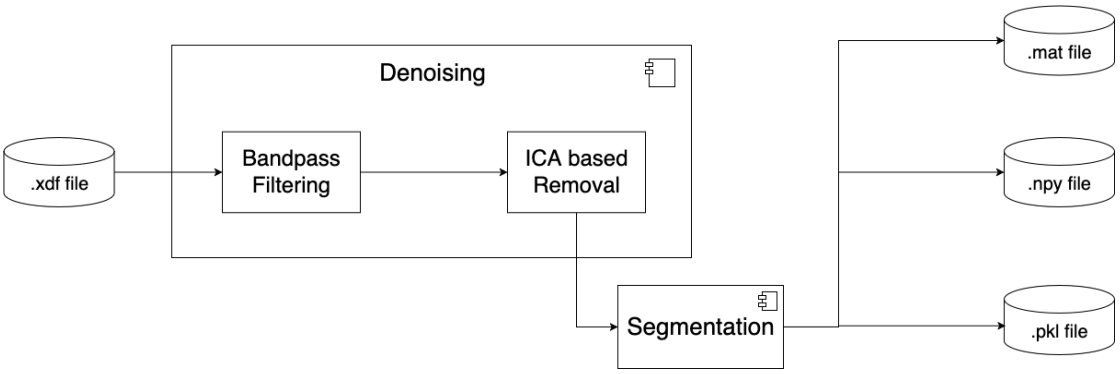


Figure 5.2: Preprocessing of the raw EEG signal

to, the signal has to be denoised, which is the most essential step. As mentioned in Section 2.4, many different artifacts besides the neural activity appear during recording. Therefore, it is crucial to recognize and omit most of these contaminations to have valuable data for feature extraction and classification. The complete preprocessing pipeline is visualized in Figure 5.2, where each step is further explained in the following.

Before explaining how the data has been denoised, it is vital to investigate the state of the raw measured data. An example can be found in Figure 5.3. Most notably, there are significant amplitude shifts over the 30-minute recording sessions. Depending on the channel, there is either an increasing or a decreasing trend in amplitude. This can be reasoned with the fact that the electrodes are DC-coupled in the system. The drift then happens due to temperature changes or skin impedance.

To overcome this signal drift, the first step of the denoising pipeline is to apply a bandpass filter on the whole session. As boundaries, 1 Hz and 30 Hz are chosen as the lower and upper borders, respectively. This handles the mentioned drift and filters out all high and low frequency noise that does not match a certain brain rhythm.

Rejection Criterion	**Threshold**
Peak-to-Peak Amplitude (PTP)	90th percentile of all components
Kurtosis	90th percentile of all components
Skewness	90th percentile of all components
Autocorrelation (lag 1)	90th percentile of all components
Relative Delta Band Power	$>$ 0.5 of total power
Relative Beta Band Power	$>$ 0.35 of total power
Mixing Matrix Sparsity	$<$ 3 active channels (relative threshold = 0.2)

Table 5.1: ICA Component Rejection Criteria

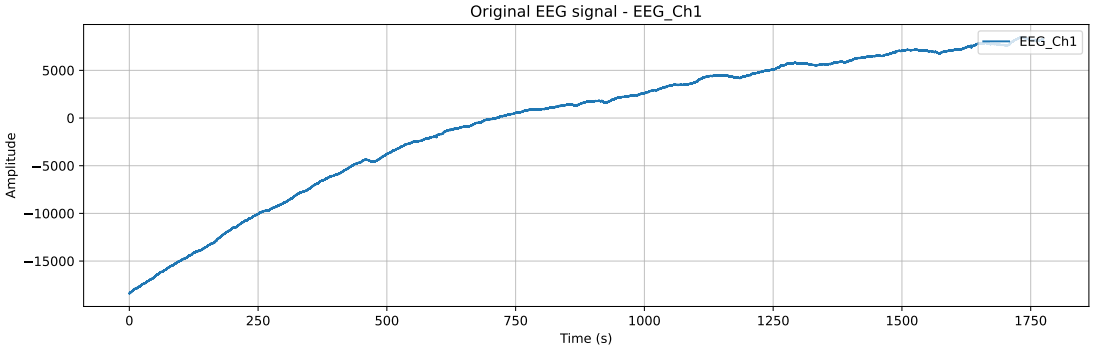


Figure 5.3: Raw EEG signal

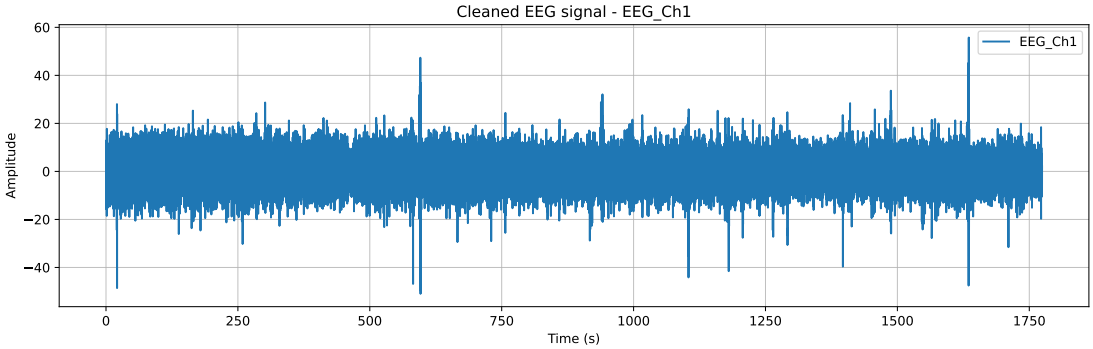


Figure 5.4: Denoised EEG signal

After removing most of the frequency-related noise, the artifacts in the EEG data must be detected and removed. An ICA algorithm is applied to the session data. Before doing that, channel three is removed from the dataset as it serves as a reference channel hence not holding valuable information. After prewhitening the data, an ICA is applied, splitting the EEG matrix into a mixing matrix accompanied by the sources. Since eight EEG channels are present, the ICA algorithm was performed with eight components. As a next step, it has to be decided whether a component is an artifact or an actual source of cerebral activity. This thesis deals with an unconventional way of measuring the EEG data, so custom criteria must be determined for this task. In Table 5.1, the rejection criteria for each component can be seen. PTP, skewness, kurtosis, and autocorrelation are used to detect unusual signal characteristics deviating from brain activity. The spectral power thresholds, including the Delta and Beta Power ratio, find artifacts by finding components dominated by specific frequency bands. Lastly, by investigating the sparsity of the mixing matrix, it is evaluated how widely a component affects the EEG channels. A component is classified as an artifact if at least two of the mentioned criteria are met. The components labelled as artifacts are then set to zero before converting the mixing matrix with the now cleaned sources to the artifact-removed version of the original EEG data. This results in a signal where most of the high amplitudes are removed with a much more consistent EEG signal, thus suggesting that most of the unwanted artifacts have been removed.

In Figure 5.4, the final denoised signal can be seen. It becomes obvious that the main drift is removed alongside with the unusual amplitudes.

With the cleaned EEG signal at hand, one can now segment the signal into its 50 individual songs[2]. Since the start and the end of one song are pushed to the LSL stream, the timestamp of those markers is read out to only keep the signal data appearing in this interval. This is done over each channel, thus resulting in a table with each song occupying a row and the channels as columns.

After all these steps, the denoised and segmented data is stored in three different formats, being .mat, .npy, and .pkl, where each session is held in one file across all file formats.

5.3 Feature extraction

The next step after the signal preprocessing is the extraction of meaningful features used for the classification task of predicting genres. Finding the right features is a difficult task in this particular use case. This is mainly because the field of MGR based

[2]The custom Python script also supports the segmentation of biosignals other than EEG. However, these will be ignored in the following as this thesis focuses on EEG signals

on EEG is pretty young, lacking in recent studies. Another problem arises from the fact that an unconventional EEG acquisition device is used with non-standardized electrode placement, on which little research exists.

To overcome those obstacles, it is essential to compute as many different features across all the domains as possible to have a versatile range of different characteristics. An optimal feature subset can be chosen using feature selection techniques, allowing for sufficient classification results.

For all features extracted from the channels, the first four seconds are taken into account. This way, the activation of the stimulus is captured.

As mentioned, features are computed across all three domains (the time domain, the frequency domain, and the time-frequency domain). All features are computed per EEG channel, thus leading to a feature space of $n_{channel} \times n_{features}$ different parameters.

An overview of all features extracted in the time domain is given in Table 5.2. The features calculated are a mix of classic statistical measures like mean, maximum, and minimum, and tailored signal metrics like entropy, RMS, and the Hjorth parameters.

Feature	**Description**
max	Maximum value of the signal
min	Minimum value of the signal
std	Standard deviation of the signal
mean	Mean value of the signal
median	Median value of the signal
var	Variance of the signal
range	Peak-to-peak range of the signal
skew	Skewness of the signal distribution
kurtosis	Kurtosis of the signal distribution
rms	Root mean square value of the signal
zcr	Zero crossing rate
h_mobility	Hjorth mobility parameter
h_complexity	Hjorth complexity parameter
energy	Total energy (sum of squares) of the signal
shannon_entropy	Shannon entropy of the signal

Table 5.2: Extracted Time-Domain Features

To gain knowledge about the signal's time properties, features from the frequency domain have also been extracted. For all features described in Table 5.3, Welch's method was applied as a basis to determine the PSD. To balance both computational

cost and accuracy, a 2 s window is used. With the PSD at hand, the spectral entropy is calculated to measure its complexity and irregularity, which can provide information about different mental states. More importantly, the relative power of each brain rhythm frequency band is calculated to further depict the neurological condition of the brain during each session. The brain rhythm's frequency bands are defined as follows:

- Delta: $0.5 - 4$ Hz
- Theta: $4 - 8$ Hz
- Alpha: $8 - 13$ Hz
- Beta: $13 - 30$ Hz

The relative power is taken to normalize the data across all sessions to account for the session-dependent measurements. In addition to the relative band power, ratios between specific rhythms can not only further normalize the results but also give a relative measure of brain state changes. This also highlights proportional changes in frequency bands, being possibly more informative than the absolute power.

Feature	**Description**
spectral_entropy	Shannon entropy of the normalized power spectral density
DELTA_relative	Relative power in the delta band
THETA_relative	Relative power in the theta band
ALPHA_relative	Relative power in the alpha band
BETA_relative	Relative power in the beta band
GAMMA_relative	Relative power in the gamma band
ALPHA_BETA_ratio	Ratio of alpha power to beta power
THETA_ALPHA_ratio	Ratio of theta power to alpha power
THETA_BETA_ratio	Ratio of theta power to beta power

Table 5.3: Extracted Frequency-Domain Features with Welch's method

To combine the time and frequency domains, a DWT calculates the features mentioned in Table 5.4. In that case, the Daubechies wavelet is used as the mother wavelet function, and a level of four is chosen, resulting in five subbands. Time domain features are extracted from each subband, including statistical measures and signal-specific parameters.

As already hinted, total features depend on the number of EEG channels, thus leading to high dimensionality. An MRMR feature selection algorithm using MID

Feature	Description
tfd_band_i_mean	Mean of wavelet coefficients in subband *i*
tfd_band_i_median	Median of wavelet coefficients in subband *i*
tfd_band_i_std	Standard deviation of wavelet coefficients in subband *i*
tfd_band_i_variance	Variance of wavelet coefficients in subband *i*
tfd_band_i_entropy	Shannon entropy of normalized wavelet energy in subband *i*
tfd_band_i_relative_energy	Relative energy of subband *i* with respect to total wavelet energy
tfd_band_i_skewness	Skewness of wavelet coefficients in subband *i*
tfd_band_i_kurtosis	Kurtosis of wavelet coefficients in subband *i*

Table 5.4: Extracted Time–Frequency Domain Features with a level 4 DWT, where *i* is the *i*-th subband

is used to choose the optimal feature set out of all calculated parameters. For both Relevance and Redundancy, the mutual information value is utilized to calculate both measures. The number of selected features is variable and can be optimized depending on the classification task or the ML model applied.

In conclusion, a rich set of versatile features provides information about many facets of the EEG signal. Furthermore, an MRMR feature selector is applied to improve the quality of the feature set during classification.

5.4 Classification

Many difficulties appear regarding classification, as it is the most challenging step in the EEG pipeline, out of two reasons specifically in this thesis. Firstly, research on EEG based MGR is minimal, where one of the only reference papers achieved mediocre performance on single EEG classification on the DEAP dataset on subject-level [12]. Secondly, the fact that the data was acquired using a multimodal EEG headset with unconventional electrode placement around the ear. Since no related work exists makes it hard to verify whether the gathered data is even suitable for such a classification task.

To overcome those hurdles, MGR is performed on the thesis's handcrafted dataset and the well-known DEAP dataset. Since this is a common source for EEG based classification and already provides genre labels provided by [12], it can be used as a source of truth in this specific classification task. Therefore, the Classification methodology can be split into two parts. The first part involves performing MGR on the DEAP dataset to research and find out which features, models and classification strategies work best. This knowledge and usable results from the first part are then transferred to the Multimodal Signals & EEG for Classification of Genres (MuSiCGen). In this way, whether the same methodology that works for a state-of-the-art dataset also applies to the MuSiCGen can be investigated, leading to faster results and a hint at the signal's quality.

5.4.1 DEAP

As already mentioned, the publicly available preprocessed DEAP dataset is used as a signal source for classification. However, the songs are not labeled according to genre; the genre tags applied by [12] are considered. In this way, comparing the results to current research in EEG based genre recognition is easily possible. This results in an imbalanced dataset because of the 40 songs, 60% belong to the POP genre and 40% to the ROCK label.

From the available 32 EEG channels, multiple features have been precomputed from each channel to depict the neural activity accordingly. The features are chosen based on past research using this dataset in emotion recognition[3]:

- Alpha, Beta, Theta, Delta, Gamma relative band power
- Alpha-Beta ratio
- Theta-Alpha ratio
- Theta-Beta ratio
- Entropy from each subband from a level 4 DWT

Important to note is that the features were calculated for the first four seconds of the recorded signal to capture the activation impulse of the stimulus. The methods used for extracting the signal are the same as mentioned in Section 2.5

To easily employ hyperparameter tuning techniques like Grid search, a modular ML pipeline is set up, illustrated in Figure 5.5. At first, a z-normalization is applied to each feature, so the data have zero mean and unit variance. This step is crucial since it

[3] cf. Section 3.4

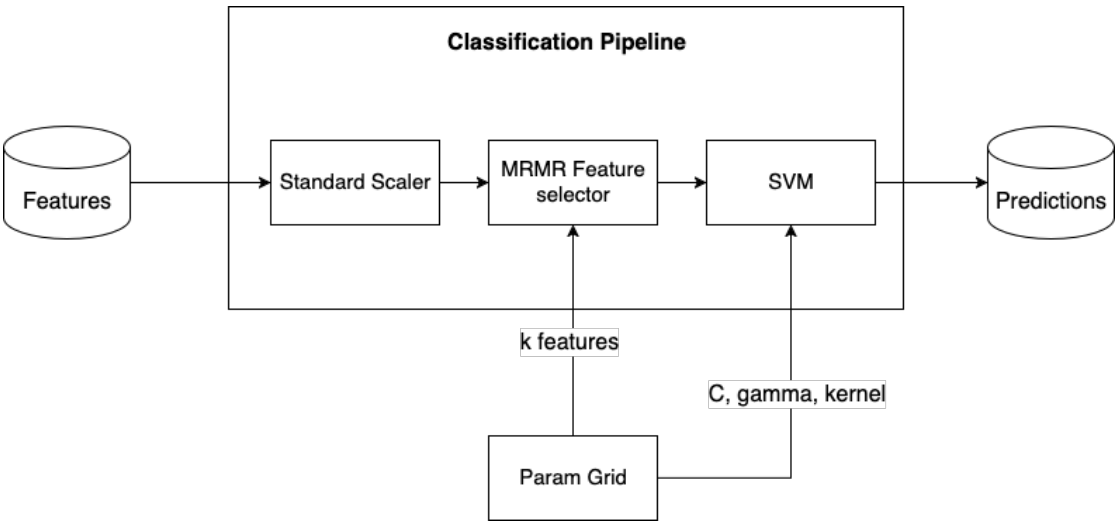


Figure 5.5: Classification Pipeline for Hyperparameter Tuning and Evaluation

accounts for abnormalities and unifies the data, leading to better classification results. An MRMR feature selection method is utilised to choose the optimal subset of features for this task. This also reduces the dimensionality, as the number of samples is over ten times less the amount of features (40×448).

The selected subset is then fed into an SVM with balanced class weight and a constant random state to make the model's outputs deterministic over multiple iterations.

In this case, a subject-based classification approach is followed. In this thesis, an individual model is optimized for each subject, allowing for a more subject-centric classification approach.

A Grid search is also embedded to find the optimal configuration to get the best-performing model possible. Apart from the SVM's hyperparameters (C, γ, kernel), the number of features selected by the MRMR is also tuned. An overview of the hyperparameter space is illustrated in Table 5.5. The distribution of the optimal parameters can then be further analysed to determine which values perform best in this dataset's task of MGR.

To evaluate the model's performance, the primary metric used is the mean accuracy with the balanced accuracy between all subjects. Alongside that, the F1-score is considered to measure both precision and recall in an imbalanced class environment. To get an idea of prediction distribution, the confusion matrices of each optimal model are also computed per subject. Moreover, the mean of those metrics over all subjects is calculated to gain information about overall performance and similarities. In the case of the confusion matrices, a global confusion matrix is generated by unifying all

predictions and test labels. Those metrics allow for a detailed analysis of MGR in the case of the DEAP dataset.

Parameter	**Search Space**
Feature count	2, 5, 8, 10, 15, 20, 30, 50
C	0.001, 0.01, 0.1, 1, 10, 100, 1000
Kernel	linear, rbf, sigmoid
gamma	scale, auto, 0.0001, 0.001, 0.01, 0.1, 1, 10

Table 5.5: Hyperparameter search space for SVM with MRMR feature selection. While feature count affects, the feature selector, the other parameters influence the SVM

5.4.2 MuSiCGen

To compare the results of the DEAP dataset with the data originating from the multi-modal EEG headset, the same methodology described in Subsection 5.4.1 is employed, including the Classification pipeline and the hyperparameter space in Figure 5.5 and Table 5.5 respectively. Nonetheless, some critical differences will be mentioned below.

Firstly, the number of channels is reduced compared to the DEAP dataset, which was measured with a professional EEG device. For this reason, and because no related work exists for the EEG headset, all features mentioned in Section 5.3 are available in the feature space from which the MRMR feature selector can choose from.

Secondly, in contrast to the DEAP dataset, the MuSiCGen is balanced in terms of genres as each session contains five songs from each possible genre (ten in total). Therefore, the balanced accuracy is not taken into account, as it will yield the same results as the mean accuracy. However, the F1-score and confusion matrices are still analysed to measure how well the model performs on multi-class prediction.

6 Results

The following chapter deals with evaluating the performance and results of the classification performed on both the DEAP dataset and the MuSiCGen set. For each dataset, the same analysis approach is followed. Firstly, the accuracies of each model are analysed alongside the F1-score for each subject/session with a focus on variance between those values. Secondly, since the hyperparameters are optimized according to each subject/session, the distribution of those parameters[1] is further investigated to see specific trends within these models. Thirdly, normalised confusion matrices are taken as a measure to get insights into the actual predictions of each model. Apart from a global confusion matrix originating from all confusion matrices combined, the best and worst classifiers' matrices are explored as well.

6.1 DEAP

Before evaluating the classification results, it is worth mentioning that the chance level[2] in this case is 50% as a binary classification is performed.

The mean and balanced accuracies of each subject are illustrated in Figure 6.1. Notably, both mean and balanced accuracy are very close to each other with 68.3% and 68.7%, respectively, concluding that each model makes overall balanced predictions. However, it is trained on an imbalanced dataset. Additionally, both the mean and balanced accuracy occupy a large range. While the mean accuracy ranges between 58% and 83%, the balanced accuracy varies from 59% to 82%. This suggests that there are many differences between subjects in EEG patterns and quality of the data. Nonetheless, each model can perform classifications above chance level, suggesting that the model actually learns patterns that are relevant for MGR in this case.

In terms of f1-score, the distribution of results follows a similar pattern to the outcomes of the accuracies, illustrated in Figure 6.2. With a mean value of 66.2% and values ranging from 55% to 82%, the findings vary across subjects. Nonetheless, since the F1-score expresses the balance between precision and recall, the models have a reasonable performance with fluctuating scores depending on the subject. Reasons for

[1] The hyperparameter space can be found in Table 5.5

[2] Accuracy expected from random guessing, calculated by the inverse of the number of classes

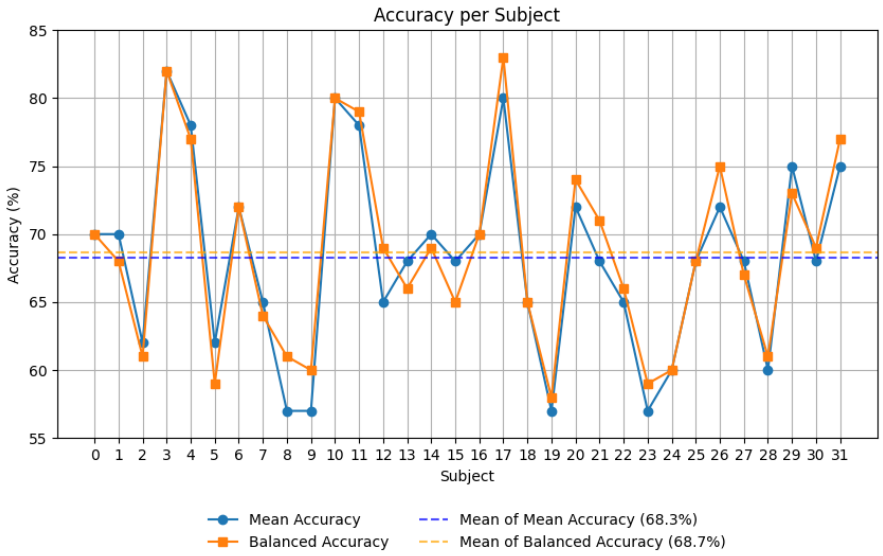


Figure 6.1: Mean and balanced accuracies for each subject's best performing model in the DEAP dataset

that could be noise or subject-dependent features.

The overall hyperparameter distribution is showcased in Figure 6.3, illustrating the count of each value appearing in an optimal configuration for the classification pipeline. It becomes obvious that the sigmoid function is the optimal kernel for most of the subjects. For C regularization, the majority of models utilize a value of 10.0, indicating the need to fit the training data more strictly for desirable results on the test validation dataset. Regarding the gamma values used, it is noticeable that the *auto* option is not in use, suggesting that it is not competitive compared to more robust options like *scale*. The absolute distribution of the feature count, however, does not give away any trend in favor of performance when looking at the overall count. Nonetheless, it is still observable that a larger count of features seems to yield better performance since most instances use a feature count greater than ten.

Figure 6.4 depicts the normalised global confusion matrix accumulated over all subjects and their respective models to get an understanding of the predictions made over all subjects. Overall, the predictions are well balanced between POP and ROCK as the predictions are right 67.3% and 69.7% respectively. Overall, it seems that the ROCK label is more predictable compared to the POP genre, although only 40% of all songs belong to this genre.

When comparing the confusion matrices of both the best and worst performing subjects, noteworthy differences can be interpreted. When looking at the best performing

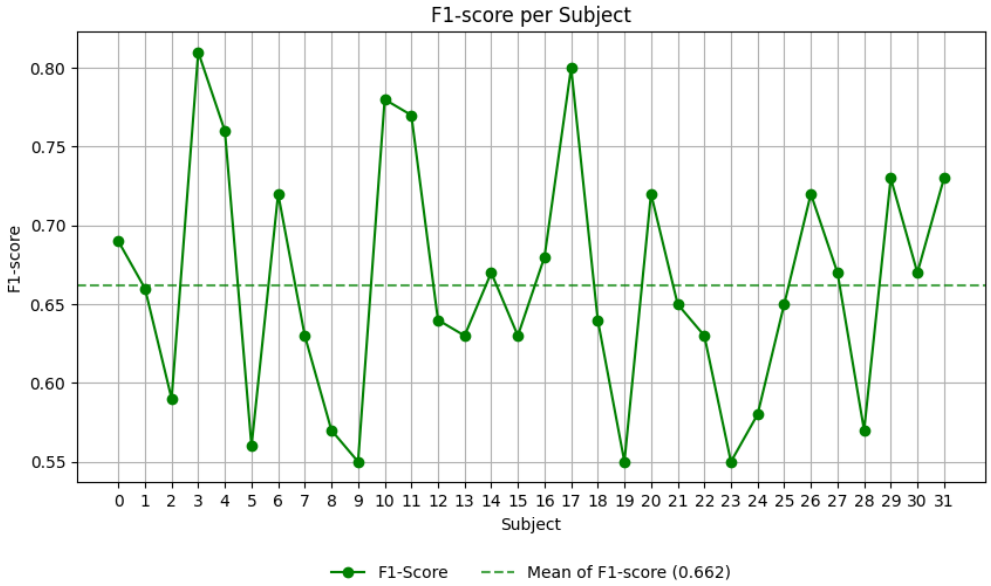


Figure 6.2: F1-scores for each subject's best performing model in the DEAP dataset

subject's confusion matrix in Figure 6.5, it becomes clear that results are very balanced and perform very well with both classes being correctly predicted with accuracies of 83% (POP) and 81% (ROCK). Contrary to that, the worst performing subject's confusion matrix, illustrated in Figure 6.6, has different attributes. Although the model predicts the ROCK songs within the average performance with 69%, the classification of the POP songs results in a chance level, indicating a bias towards the ROCK genre. Setting both matrices side to side emphasizes the subject-dependent patterns in neural activity being different and unique per person.

6.2 MuSiCGen

After analysing the MGR results on a well-known dataset, the classification results on the MuSiCGen are investigated following the same structure as in Section 6.1. In this case, a ten-class classification task is performed on the MuSiCGen dataset leading to a chance level of 10%.

When looking at the mean accuracies of each model in Figure 6.7, similar properties like in the DEAP dataset can be observed. Overall, each session is classified above chance level since the predictions are made over ten genres instead of just two. With a mean accuracy of 20.2% across all sessions, a minimum accuracy of 16% and a

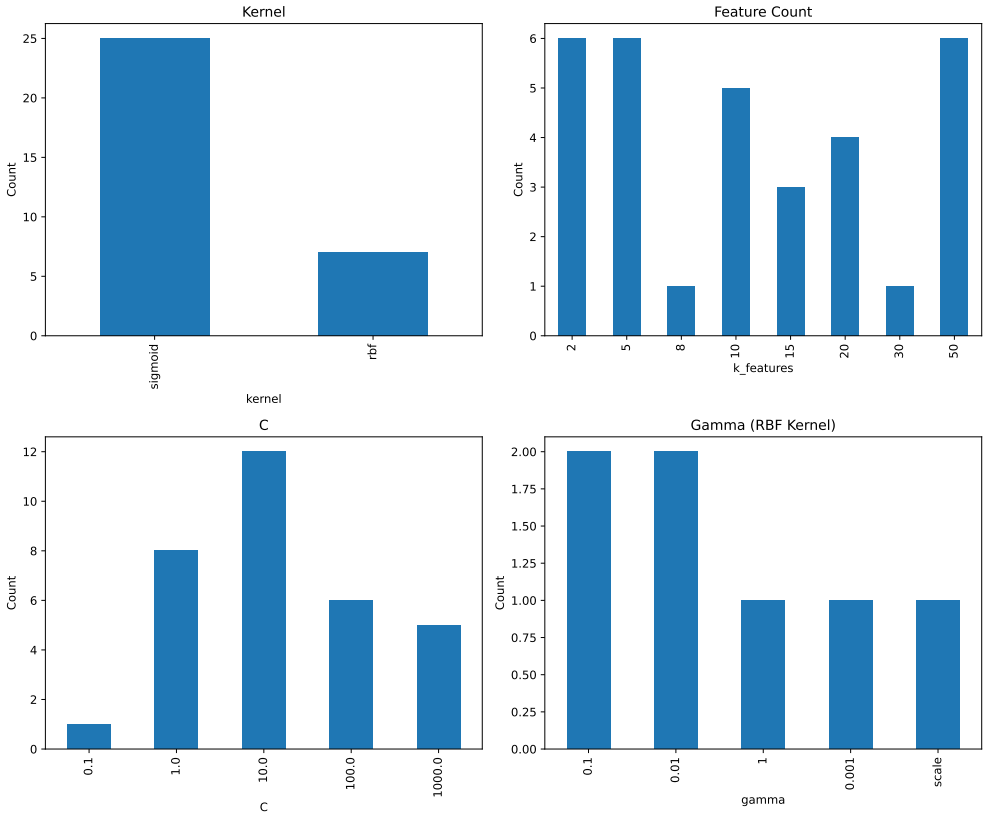


Figure 6.3: The distribution of optimal hyperparameters across all subjects in the DEAP dataset presented in absolute counts

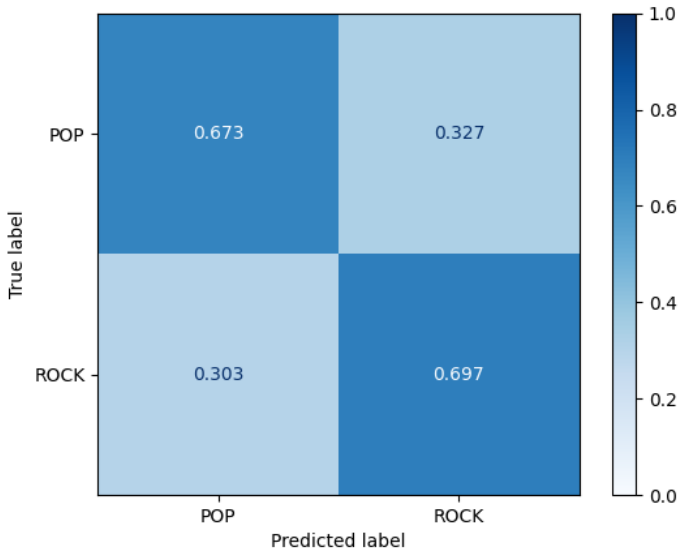


Figure 6.4: Global confusion matrix accumulated over all subjects of the DEAP dataset

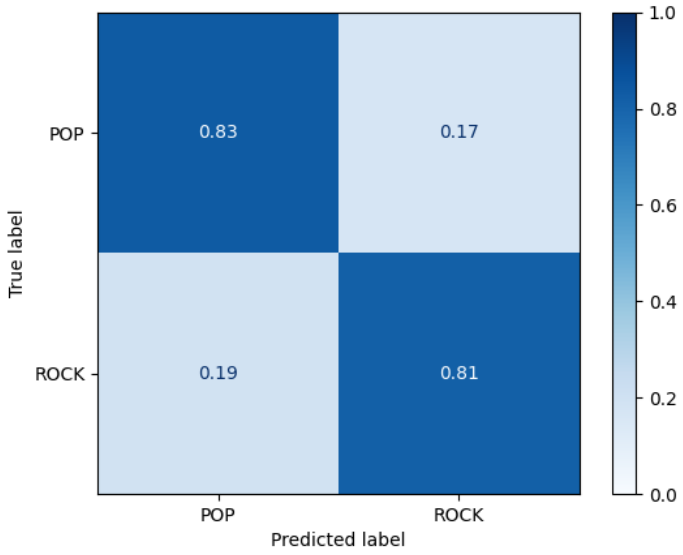


Figure 6.5: Confusion matrix of the best performing subject in the DEAP dataset (Subject 3, F1-score of 0.83)

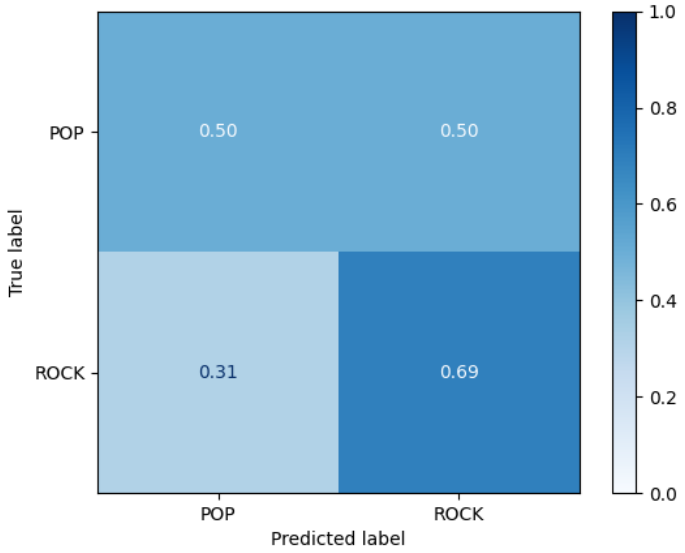


Figure 6.6: Confusion matrix of the worst performing subject in the DEAP dataset (Subject 9, F1-score of 0.55)

maximum accuracy of 26%, there is also a high variance between subjects, but not as large as the one in the DEAP dataset. Nonetheless, performance highly fluctuates between sessions. One reason for that could be that the headset's electrodes had slightly different placement across sessions, influencing the quality of the EEG signal. Another reason may be that some songs evoke clearer genre-specific states in the brain, so classification becomes easier for those music pieces. Apart from that, environmental conditions or accidental head movements can be influencing factors for this alteration of accuracy.

The F1-scores, on the other hand, are shown in Figure 6.8 and show a mean of 14.7%, which is only slightly better than chance level. The fluctuation also remains consistent with a minimum of 9% and a maximum of 24%. This result implies that the models do not capture structures of each genre equally, indicating the difficulty in multi-class prediction in EEG based MGR.

The optimal hyperparameter space is displayed in Figure 6.9. Similar to the DEAP dataset, the sigmoid kernel is utilized the most, pointing out that the sigmoid kernel seems to be a favourable kernel for finding patterns in EEG data for MGR. Unlike the properties of the kernel, trends in feature count are different compared to the DEAP. While the classification on the DEAP data seems to favor a larger amount of features

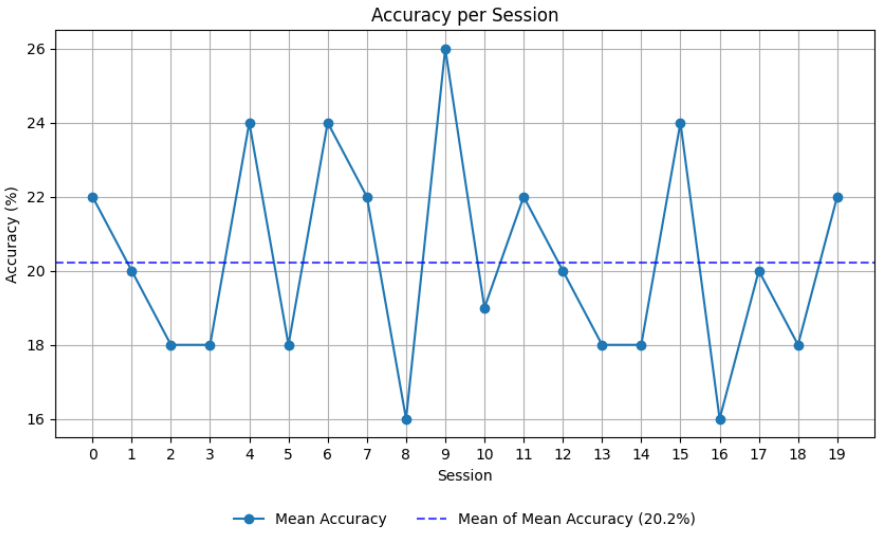


Figure 6.7: Mean accuracies of each session's best performing models in the MuSiCGen

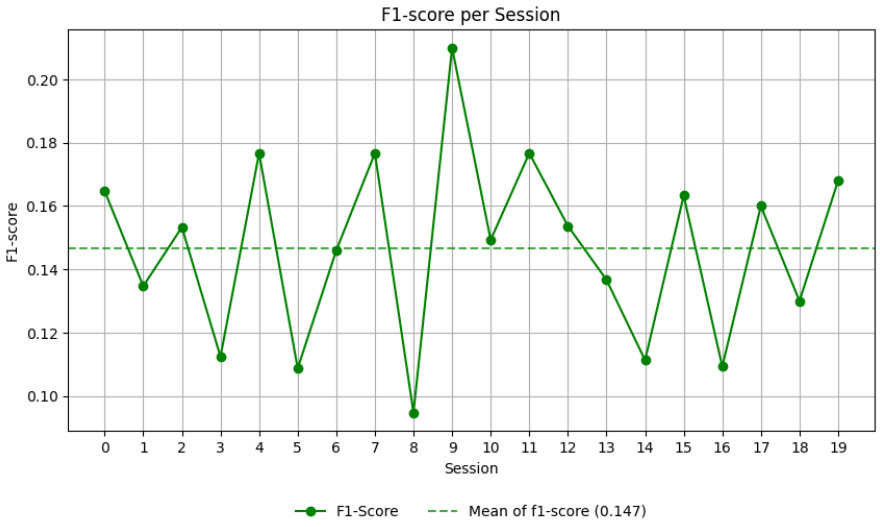


Figure 6.8: Mean f1-scores of each session's best performing models in the MuSiCGen

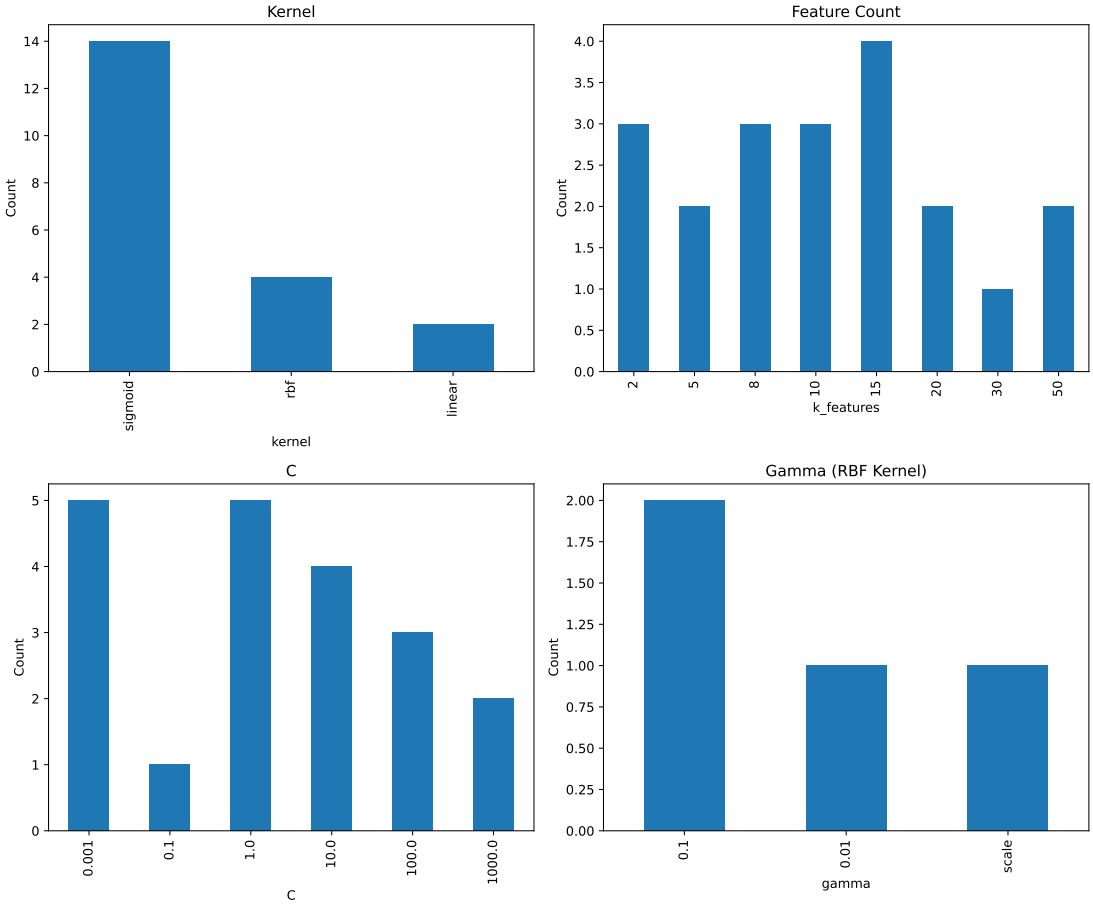


Figure 6.9: The distribution of optimal hyperparameters across all sessions in the MuSiCGen presented in absolute counts

(more than ten), the models of the MuSiCGen prefer a lower amount of features (less than 15). The distribution of the C value is also different compared to the DEAP data. In the case of the MuSiCGen, low and high C values come up nearly with the same frequency, with 0.001 and 1.0 being used most frequently. This smoother decision boundary by the low C values could be explained by the overlapping of genres or general noise in the EEG data.

As already suggested by the low F1-score, not all classes are predicted equally well, which is emphasized by the overall confusion matrix in Figure 6.10. The classes predicted correctly the most are blues, country, and disco with 27%, while metal was the worst with only 12%. It is also interesting to see that some genres get mistaken for other genres. For instance, classical music was predicted correctly only 15% of the time and was mistaken as country music 17% of the time. Nonetheless, there are no very large misclassifications in genres, which gives the hint that it either is difficult to distinguish between ten music genres based on EEG or that an SVM is not able to capture genre-specific patterns well.

Overall, there lies some information about different genres in the headset's acquired signal. However, since the performance is only a little bit above chance level, it raises concerns about the data quality of this multimodal approach. However, the results of the DEAP dataset are not much better than the MuSiCGen, even though it is a binary decision, hinting that multi-class MGR based on EEG signals is still a difficult task for SVMs, requiring more research in the future.

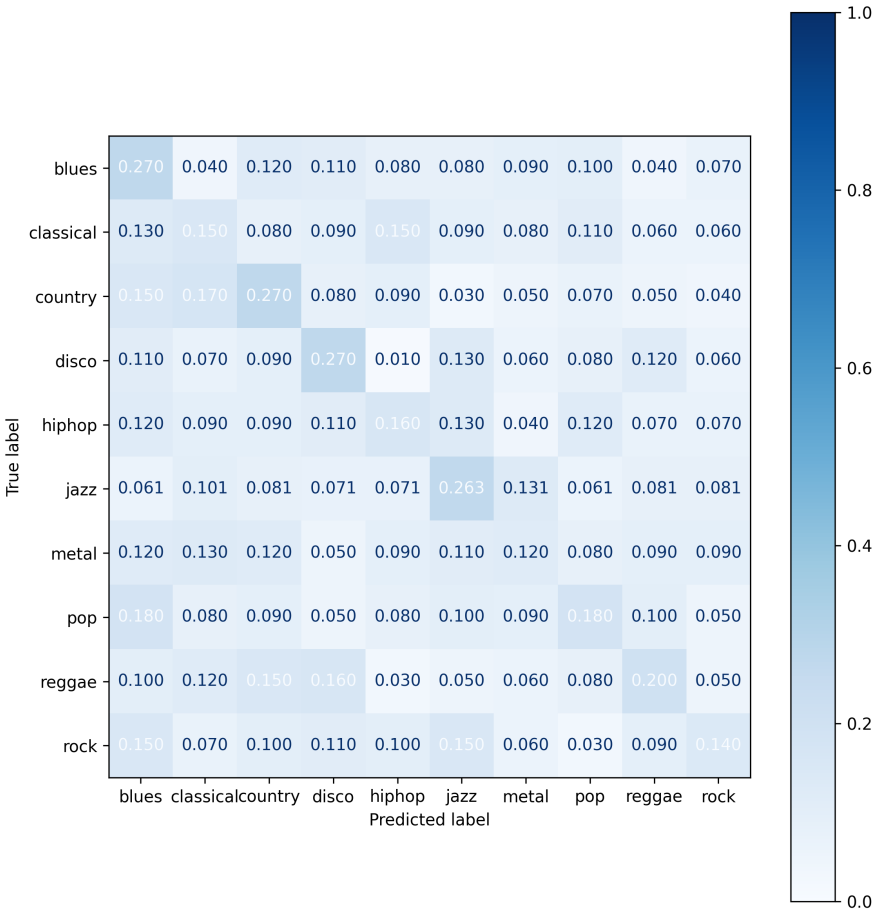


Figure 6.10: Global confusion matrix accumulated over all sessions of the MuSiCGen dataset

7 Conclusion

7.1 Key Findings and Results

This thesis provides multiple results and findings relevant to the research area of EEG based classification for MGR. For these purposes, a novel EEG dataset called MuSiCGen tailored for MGR is introduced, being the first of its kind. The balanced dataset consisting of 100 songs per genre, with ten genres, is available in standard file formats and is easily reproducible. Furthermore, a multimodal EEG headset was utilised to measure the signal, fostering novel approaches that could make their way into everyday life. As the headset also acquires IMU, PPG and temperature than the neural activity, those were also measured and are accessible in MuSiCGen.

Going hand in hand with MuSiCGen dataset, a denoising and preprocessing pipeline is presented that works on fewer electrodes and unconventional placement of those. By using an ICA based approach combined with a score-based rejection mechanism, it is possible to detect artifacts in the measured signal to eliminate them from the data.

Apart from the creation of the MuSiCGen dataset, MGR has been performed not only on those data but also on the well-known DEAP dataset to have another source of brain signals for comparison. Overall, it is possible to achieve accuracy and F1-scores that are better than chance level on classification for each subject independently. This infers that there are neural patterns specific to musical genres that can be captured by ML methods, being a promising finding for further research. Moreover, a fluctuating trend and variance in performance metrics are observable, leading to the conclusion that there are inconsistencies in music genre patterns and data quality across sessions for the DEAP and the MuSiCGen, respectively. Nonetheless, it is significant that most optimal models use a sigmoid kernel, implying that this function can capture most of the genre-specific patterns.

7.2 Limitations and Future Work

Although this thesis provides key findings and conclusions for the area of EEG based MGR, some problems and limitations are still isolated from this thesis. The first one lies in the multimodal EEG headset approach, as it remains questionable whether the

placement of electrodes around the ear can measure the neural activity well. This concern is because state-of-the-art EEG devices place the electrodes on top of the scalp with a large number (mainly 32-64) of electrodes. The headset, however, has fewer electrodes (seven) and is not placed on the scalp but around the ear.

Another limitation is that each model is optimized and trained for each subject/session individually. This strategy may work for tasks that require dependency on the subject (e. g. personalised therapy), but does not give away information about a possible general purpose model being able to perform well on unseen subjects.

Even though the performances are above chance level, the overall results are still only slightly better than this. Multiclass classification for the MuSiCGen remains difficult, especially as the differentiation between more than two genres seems hard to capture.

To overcome the mentioned limitations and problems, further research in the area of EEG based MGR has to be conducted. First, to verify the quality of the EEG signal recorded by the headset, it would be vital to record the neural activity with a state-of-the-art EEG device and standardised denoising steps on the same songs and session setting. This way, it would be possible to compare both signals with each other to make statements about the quality of the signal of the multimodal headset.

Apart from comparing the headset's signal with data originating from standard acquisition methods, it is also important to further research on the concept of such headphones. For instance, electrodes could be added on the headband of the headset. This not only increases the amount of EEG channels but also adds measurement units on the scalp, mimicking state-of-the-art EEG systems. Another idea would be to include an EEG headcap with the headset by installing it on the headphone's headband. This way the convenience of the bluetooth headset is combined with the measurement quality of the EEG headcap.

Moreover, extending the MuSiCGen dataset further would be beneficial in providing more versatile data. Besides adding more songs, different subjects could be recorded. For example, seeing how the signal differs between musicians and non-musicians would be interesting.

Other future work consists of a different classification evaluation by optimising per subject/session and for all data collectively. This way, more evidence can be given on how well ML methods can generalise on different subjects and/or different environments. Combined with that, it would be interesting to see how well different ML techniques perform on this task. Deep Learning architectures have the potential to yield even better results as they are widely used in other applications regarding EEG.

Lastly, since the headset also measures other biosignals, those can be used to train multi-modal models. It can help capture artifacts better and offer valuable information that can potentially improve performance.

Abbreviations

ADC Analogue to Digital converter

AP Action potential

BCI Brain-Computer Interfacing

BMI Brain Machine Interface

BSS Blind Source Seperation

CNN Convolutional Neural Network

CNS Central Nervous system

CWT Continous Wavelet Transform

DEAP Database for Emotion Analysis using Physiological Signals

DFT Discrete Fourier Transform

DL Deep Learning

DWT Discrete Wavelet Transform

ECG Electrocardiography

EEG Electroencephalography

EMG Electromyography

EOG Electrooculography

fMRI functional Magnetic resonsance imaging

FMA Free Music Archive

GMM Gaussian Mixture Model

HOS Higher-order statistics

ICA Independent Component Analysis

IMU Inertial Measurement Unit

DL Deep Learning

LR Logistic Regression

LSL Lab Streaming Layer

MARA Multiple Artifact Rejection Algorithm

MFCC Mel Frequency Cepstral Coefficients

MGR Music Genre Recognition

MIR Music Information Retrieval

ML Machine Learning

MRMR Minimum Redundancy Maximum Relevance

MuSiCGen Multimodal Signals & EEG for Classification of Genres

PCA Principal Component Analysis

PPG Photoplethysmogram

PSD Power Spectral Density

PTP Peak-to-Peak Amplitude

RMS Root Mean Square

SEED SJTU Emotion EEG Dataset

STFT Short Time Fourier Transform

SVM Support Vector Machine

TFA Time-frequency Analysis

XGB Extreme Gradient Boosting

ZCR Zero Crossing Rate

K-NN K-neares neighbor

List of Figures

2.1 The five brain rhythms in EEG [28] 6
2.2 Conventional 10-20 EEG electrode positions for 21 electrodes [28] 7
2.3 Comparision between Time and Frequency domain. From the signal (on the right), sinusoids (on the left) are extracted that have unique frequencies and amplitudes (at the bottom) [17] 14
2.4 Visualization of a DWT [10] 15

5.1 Technical setup 33
5.2 Preprocessing of the raw EEG signal 34
5.3 Raw EEG signal 35
5.4 Denoised EEG signal 35
5.5 Classification Pipeline for Hyperparameter Tuning and Evaluation . . . 41

6.1 Mean and balanced accuracies for each subject's best performing model in the DEAP dataset 44
6.2 F1-scores for each subject's best performing model in the DEAP dataset 45
6.3 The distribution of optimal hyperparameters across all subjects in the DEAP dataset presented in absolute counts 46
6.4 Global confusion matrix accumulated over all subjects of the DEAP dataset 47
6.5 Confusion matrix of the best performing subject in the DEAP dataset (Subject 3, F1-score of 0.83) 47
6.6 Confusion matrix of the worst performing subject in the DEAP dataset (Subject 9, F1-score of 0.55) 48
6.7 Mean accuracies of each session's best performing models in the MuSiCGen 49
6.8 Mean f1-scores of each session's best performing models in the MuSiCGen 49
6.9 The distribution of optimal hyperparameters across all sessions in the MuSiCGen presented in absolute counts 50
6.10 Global confusion matrix accumulated over all sessions of the MuSiCGen dataset 52

List of Tables

2.1 Types of artifacts in EEG signals [17] 8

4.1 Overview of potential Music datasets. The attributes only correspond to the parts of the dataset that are interesting in terms of MGR. The term undefined is used when there are no clear genre tags or if they depend on subjective interpretation and individual preprocessing 29

5.1 ICA Component Rejection Criteria 34
5.2 Extracted Time-Domain Features 37
5.3 Extracted Frequency-Domain Features with Welch's method 38
5.4 Extracted Time–Frequency Domain Features with a level 4 DWT, where i is the i-th subband 39
5.5 Hyperparameter search space for SVM with MRMR feature selection. While feature count affects, the feature selector, the other parameters influence the SVM 42

Bibliography

[1] S. Abdumalikov, J. Kim, and Y. Yoon. "Performance analysis and improvement of machine learning with various feature selection methods for EEG-based emotion classification." In: *Applied Sciences* 14.22 (2024), p. 10511.

[2] R. Alhalaseh and S. Alasasfeh. "Machine-learning-based emotion recognition system using EEG signals." In: *Computers* 9.4 (2020), p. 95.

[3] H. Bahuleyan. "Music Genre Classification using Machine Learning Techniques." In: *CoRR* abs/1804.01149 (2018). arXiv: 1804.01149.

[4] B. Bein. "Entropy." In: *Best Practice & Research Clinical Anaesthesiology* 20.1 (2006). Monitoring Consciousness, pp. 101–109. ISSN: 1521-6896. DOI: https://doi.org/10.1016/j.bpa.2005.07.009.

[5] T. Bertin-Mahieux, D. P. Ellis, B. Whitman, and P. Lamere. "The million song dataset." In: *Ismir*. Vol. 2. 9. 2011, p. 10.

[6] H. Candra, M. Yuwono, R. Chai, A. Handojoseno, I. Elamvazuthi, H. T. Nguyen, and S. Su. "Investigation of window size in classification of EEG-emotion signal with wavelet entropy and support vector machine." In: *2015 37th Annual international conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. IEEE. 2015, pp. 7250–7253.

[7] S. Chatterjee and Y.-C. Byun. "EEG-based emotion classification using stacking ensemble approach." In: *Sensors* 22.21 (2022), p. 8550.

[8] X. Cui, Y. Wu, J. Wu, Z. You, J. Xiahou, and M. Ouyang. "A review: Music-emotion recognition and analysis based on EEG signals." In: *Frontiers in neuroinformatics* 16 (2022), p. 997282.

[9] M. Defferrard, K. Benzi, P. Vandergheynst, and X. Bresson. "FMA: A dataset for music analysis." In: *arXiv preprint arXiv:1612.01840* (2016).

[10] A. S. Al-Fahoum and A. A. Al-Fraihat. "Methods of EEG signal features extraction using linear analysis in frequency and time-frequency domains." In: *International Scholarly Research Notices* 2014.1 (2014), p. 730218.

[11] J. F. Gemmeke, D. P. Ellis, D. Freedman, A. Jansen, W. Lawrence, R. C. Moore, M. Plakal, and M. Ritter. "Audio set: An ontology and human-labeled dataset for audio events." In: *2017 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE. 2017, pp. 776–780.

[12] P. Ghaemmaghami and N. Sebe. "Brain and music: Music genre classification using brain signals." In: *2016 24th European Signal Processing Conference (EUSIPCO)*. IEEE. 2016, pp. 708–712.

[13] R. M. Gray. *Entropy and information theory*. Springer Science & Business Media, 2011.

[14] U. Güçlü, J. Thielen, M. Hanke, and M. A. J. van Gerven. *Brains on Beats*. 2016. arXiv: 1606.02627 [q-bio.NC].

[15] M. Haggblade, Y. Hong, and K. Kao. "Music genre classification." In: *Department of Computer Science, Stanford University* (2011).

[16] R. Hasan and S. M. R. Islam. "A Comparative Analysis of Emotion Recognition from EEG Signals Using Temporal Features and Hyperparameter-Tuned Machine Learning Techniques." In: *MethodsX* (2025), p. 103468.

[17] L. Hu and Z. Zhang. *EEG signal processing and feature extraction*. Springer, 2019.

[18] I. Jo, S. Lee, and S. Oh. "Improved measures of redundancy and relevance for mRMR feature selection." In: *Computers* 8.2 (2019), p. 42.

[19] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras. "DEAP: A Database for Emotion Analysis ;Using Physiological Signals." In: *IEEE Transactions on Affective Computing* 3.1 (2012), pp. 18–31. DOI: 10.1109/T-AFFC.2011.15.

[20] J. S. Kumar and P. Bhuvaneswari. "Analysis of electroencephalography (EEG) signals and its categorization–a study." In: *Procedia engineering* 38 (2012), pp. 2525–2536.

[21] E. Law, K. West, M. I. Mandel, M. Bay, and J. S. Downie. "Evaluation of algorithms using games: The case of music tagging." In: *ISMIR*. Citeseer. 2009, pp. 387–392.

[22] X. Li, D. Song, P. Zhang, Y. Zhang, Y. Hou, and B. Hu. "Exploring EEG features in cross-subject emotion recognition." In: *Frontiers in neuroscience* 12 (2018), p. 162.

[23] D. Liang, H. Gu, and B. O'Connor. "Music genre classification with the million song dataset." In: *Machine Learning Department, CMU* (2011).

[24] Y.-P. Lin, C.-H. Wang, T.-P. Jung, T.-L. Wu, S.-K. Jeng, J.-R. Duann, and J.-H. Chen. "EEG-based emotion recognition in music listening." In: *IEEE Transactions on Biomedical Engineering* 57.7 (2010), pp. 1798–1806.

[25] K. P. Miyapuram, N. Ahmad, P. Pandey, and J. D. Lomas. "Electroencephalography (EEG) dataset during naturalistic music listening comprising different genres with familiarity and enjoyment ratings." In: *Data in Brief* 45 (2022), p. 108663. ISSN: 2352-3409. DOI: https://doi.org/10.1016/j.dib.2022.108663.

[26] E. Moulin. "RMS Calculation for Energy Meter Applications Using the ADE7756." In: *Analog Devices* (2003).

[27] L. Rotheneder. "An explanation system for CNN music genre classification based on semantic descriptorsn." PhD thesis. Technische Universität Wien, 2023.

[28] S. Sanei and J. A. Chambers. *EEG signal processing and machine learning*. John Wiley & Sons, 2021.

[29] A. K. Singh and S. Krishnan. "Trends in EEG signal feature extraction applications." In: *Frontiers in artificial intelligence* 5 (2023), p. 1072801.

[30] M. S. Treder, H. Purwins, D. Miklody, I. Sturm, and B. Blankertz. "Decoding auditory attention to instruments in polyphonic music using single-trial EEG classification." In: *Journal of Neural Engineering* 11.2 (Mar. 2014), p. 026009. DOI: 10.1088/1741-2560/11/2/026009.

[31] G. Tzanetakis and P. Cook. "Musical genre classification of audio signals." In: *IEEE Transactions on speech and audio processing* 10.5 (2002), pp. 293–302.

[32] J. A. Urigüen and B. Garcia-Zapirain. "EEG artifact removal—state-of-the-art and guidelines." In: *Journal of Neural Engineering* 12.3 (Apr. 2015), p. 031001. DOI: 10.1088/1741-2560/12/3/031001.

[33] C. Vidaurre, N. Krämer, B. Blankertz, and A. Schlögl. "Time domain parameters as a feature for EEG-based brain–computer interfaces." In: *Neural Networks* 22.9 (2009), pp. 1313–1319.

[34] X.-W. Wang, D. Nie, and B.-L. Lu. "Emotional state classification from EEG data using machine learning approach." In: *Neurocomputing* 129 (2014), pp. 94–106.